MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION, MUMBAI

A Project Report on

Web Application for Water Jar Distribution System

Submitted as a partial fulfillment for the award of

Diploma in Computer Engineering

Submitted by

Harshada Pandurang Mahajan

Khushbu Ravindra Patil

Tanishka Hemant Patil

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DEPARTMENT OF COMPUTER ENGINEERING

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(Institute Code : 0059)

2025 – 2026

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Khushbu Ravindra Patil	23610960248
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Guided by.

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DEPARTMET OF COMPUTER ENGINEERING

CERTIFICATE

This is to certify that,

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has satisfactorily completed project entitled "**Web Application for Water Jar Distribution System**" in partial fulfillment for the **Diploma Course in Computer Engineering** of the **Maharashtra State Board of Technical Education** at our polytechnic during the academic year **2025-2026**. The project is completed by a group consisting of four persons under the guidance of the Faculty Guide.

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Place: Dhule

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Faculty Guide

Signature:

Name :

External Examiner

Signature:

Name : Mr. N. D.Patel

H.O.D.

Signature:

Name : Prof. Sanjay Hire

Principal.

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TO WHOMSOEVER IT MAY CONCERN

This is to certify that the following students of **Final Year Diploma in Computer Engineering** from **S.S.V.P.S's B. S. Deore Polytechnic, Dhule** have successfully completed their **Capstone Project** titled: **"Web Application for Water Jar Distribution System"**

Name of Students:

1. Harshada Pandurang Mahajan.
2. Khushbu Ravindra Patil.
3. Tanishka Hemant Patil.
4. Priti Tukaram Patil.

This project was sponsored by our organization and developed as per our requirements.

We appreciate their sincere efforts, dedication, and hard work during the project.

We wish them all the best for their bright future.

For,
Gurumauli Water Jar Plan,Gulab Bhau Nagar,Mohadi,Dist.Dhule-424311

Acknowledgement

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Finally, we are truly thankful to our families and friends for their constant encouragement, motivation, and support throughout the project. Their support has been very helpful in completing our work successfully.

Harshada Pandurng Mahajan

Khushbu Ravindra Patil

Tanishka Hemant Patil

Priti Tukaram Patil

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Abstract

The Online Water Jar Supplier Management System is a website created for a water jar supplier to manage customer orders, deliveries, and payment records in a simple and organized way. Currently, the supplier maintains all information manually in notebooks or through phone calls, which can lead to mistakes, missing records, and difficulty in tracking daily, weekly, or monthly orders. This system solves these problems by keeping all customer details, order information, delivery status, and payment records in one place digitally. Customers can easily place orders online, specifying the number of jars and delivery address, while the supplier can see all orders clearly, update deliveries, and keep proper records automatically. The system also helps to check which customers are regular, how many jars were delivered in a day or week, and how the business is running over time. By using this system, manual work is reduced, mistakes are avoided, and managing the water jar business becomes much easier and faster.

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CHAPTER 1
INTRODUCTION

Water is one of the most essential needs of daily life, and many households and businesses depend on packaged drinking water supplied in large jars. In local areas, water jar suppliers play an important role in delivering clean drinking water to customers on a regular basis. These suppliers usually manage orders, deliveries, and payments manually using notebooks or paper records.

In the traditional system, the supplier writes down details such as customer name, number of jars ordered, delivery date, and payment status. While this method works for small operations, it becomes difficult to manage as the number of customers increases. Manual record-keeping can lead to errors, data loss, and confusion in tracking pending orders and payments. It is also time-consuming and inefficient to maintain daily, weekly, and monthly records.

To overcome these problems, there is a need for a digital solution that can simplify and organize the entire process. A web-based system can help the supplier manage all activities in one place, reduce manual work, and improve accuracy. It also allows better tracking of customer orders, delivery status, and payment records.

The "Web Application for Water Jar Distribution System" is developed to provide an efficient and user-friendly platform for managing water jar supply operations. This system enables customers to place orders online and allows the supplier to monitor and control all aspects of the business digitally. By using this application, the supplier can save time, reduce errors, and improve overall service quality.

This project aims to modernize the traditional water jar distribution process by introducing a simple and effective web-based solution.

In the manual water jar distribution system, all the activities are performed manually by the supplier. The supplier maintains records in notebooks or registers without using any digital system. Important details such as customer name, order quantity, delivery date, and payment status are written by hand. This method is simple but becomes difficult to manage as the number of customers increases. There is a high chance of human error, missing data, and confusion while handling multiple orders. Tracking pending deliveries and payments is also not easy in this system.

- **Existing Manual System Workflow**

1. **Customer Places Order**

   In the traditional system, the customer places an order by calling the supplier or visiting the shop personally. There is no online system available, so all communication is done manually. The customer tells the supplier how many water jars are needed and sometimes also mentions the delivery time.

2. **Writing Order Details**

   After receiving the order, the supplier writes all the details in a notebook. This includes the customer's name, contact number, number of jars ordered, and date of order. Since this process is done manually, there is a possibility of mistakes or missing information.

3. **Planning Delivery**

   The supplier then plans the delivery based on the orders received. This planning is done manually without any system support. The supplier decides which orders should be delivered first and assigns the delivery work accordingly. This can become confusing when there are many orders.

4. **Delivery of Water Jars**

   The delivery person collects the required number of water jars and delivers them to the customer's address. The delivery is done based on the supplier's instructions. There is no digital tracking system, so the process depends completely on manual communication.

5. **Updating Delivery Status**

   Once the delivery is completed, the supplier updates the status in the notebook. The supplier may put a tick mark or write a note to indicate that the order has been delivered. If the update is missed, it may cause confusion later.

6. **Recording Payment**

   The supplier records the payment details in a separate notebook or in the same register. If the customer pays immediately, it is marked as paid using a tick or note. For regular customers, payments may be recorded weekly or monthly. This manual tracking can lead to errors or missed entries.

7. **Checking Records**

   To review business activity, the supplier checks the notebook manually. The supplier goes through pages to track daily, weekly, or monthly orders and payments. This process takes time and becomes difficult when records increase.

CHAPTER 2

LITERATURE SURVEY

This chapter presents the study of existing water jar distribution systems used by local suppliers. It explains how the current system works, mainly using manual methods like notebooks and phone calls to manage orders, deliveries, and payments. In most cases, these traditional systems are simple but not efficient. It becomes difficult to manage records properly when the number of customers increases. There is no proper way to track orders, delivery status, or payment details.

This chapter also highlights the problems in the existing system, such as errors in records, difficulty in tracking, and time-consuming processes. These issues reduce efficiency and create confusion. The purpose of this study is to understand these problems and identify the need for a better and more organized digital system.

2.1 REQUIREMENT GATHERING

Questions Asked to Water Jar Supplier:

Q1. Do you face problems in managing orders manually?

Answer: Yes, it becomes difficult to manage all orders using notebooks, especially when there are many customers.

Q2. Is it difficult to track daily, weekly, or monthly orders?

Answer: Yes, it takes time to check records and calculate orders manually.

Q3. Do you face issues in tracking pending deliveries?

Answer: Yes, sometimes it is confusing to know which orders are delivered and which are still pending.

Q4. Do you find it difficult to manage customer details?

Answer: Yes, maintaining customer information in notebooks is time-consuming and not well organized.

Q5. Do you face problems while managing payment records?

Answer: Yes, it is difficult to track which customers have paid and which payments are still pending.

Q6. Is there any confusion in maintaining regular customer payments (weekly/monthly)?

Answer: Yes, sometimes it becomes confusing to track regular payments manually.

Q7. Do you experience errors in manual record keeping?

Answer: Yes, there are chances of mistakes while writing or updating records.

Q8. Is it difficult to find old records when needed?

Answer: Yes, searching previous records in notebooks takes a lot of time.

Q9. Do you have any system to track delivery staff work?

Answer: No, delivery work is managed manually without proper tracking.

Q10. Do you think a digital system will help improve your work?

Answer: Yes, a digital system will make it easier to manage orders, deliveries, and payments efficiently.

The water jar supplier faces many problems while managing work manually. It is difficult to handle orders, deliveries, and payment records using notebooks. Sometimes there is confusion in tracking which orders are completed and which are still pending. Managing daily, weekly, and monthly records also takes a lot of time. There are chances of mistakes and missing data. Because there is no digital system, the work becomes slow and less efficient. So, there is a need for a system that can make the work easy, save time, and manage all records properly.

2.2 STUDY OF EXISTING SYSTEM

This section presents the study of the existing water jar distribution system used by the supplier. The system mainly includes manual methods and basic communication like phone calls for managing orders, deliveries, and payments.

2.2.1 Manual Process System

Currently, the water jar supplier uses a manual system to manage the work. The supplier writes order details, customer information, delivery records, and payment details in notebooks or registers. Customers usually place orders by calling or visiting.

This system depends on manual work, so it takes more time and effort. It becomes difficult to manage when the number of customers increases. There are chances of mistakes, missing records, and confusion in tracking orders and payments. Because of this, the system is not very efficient.

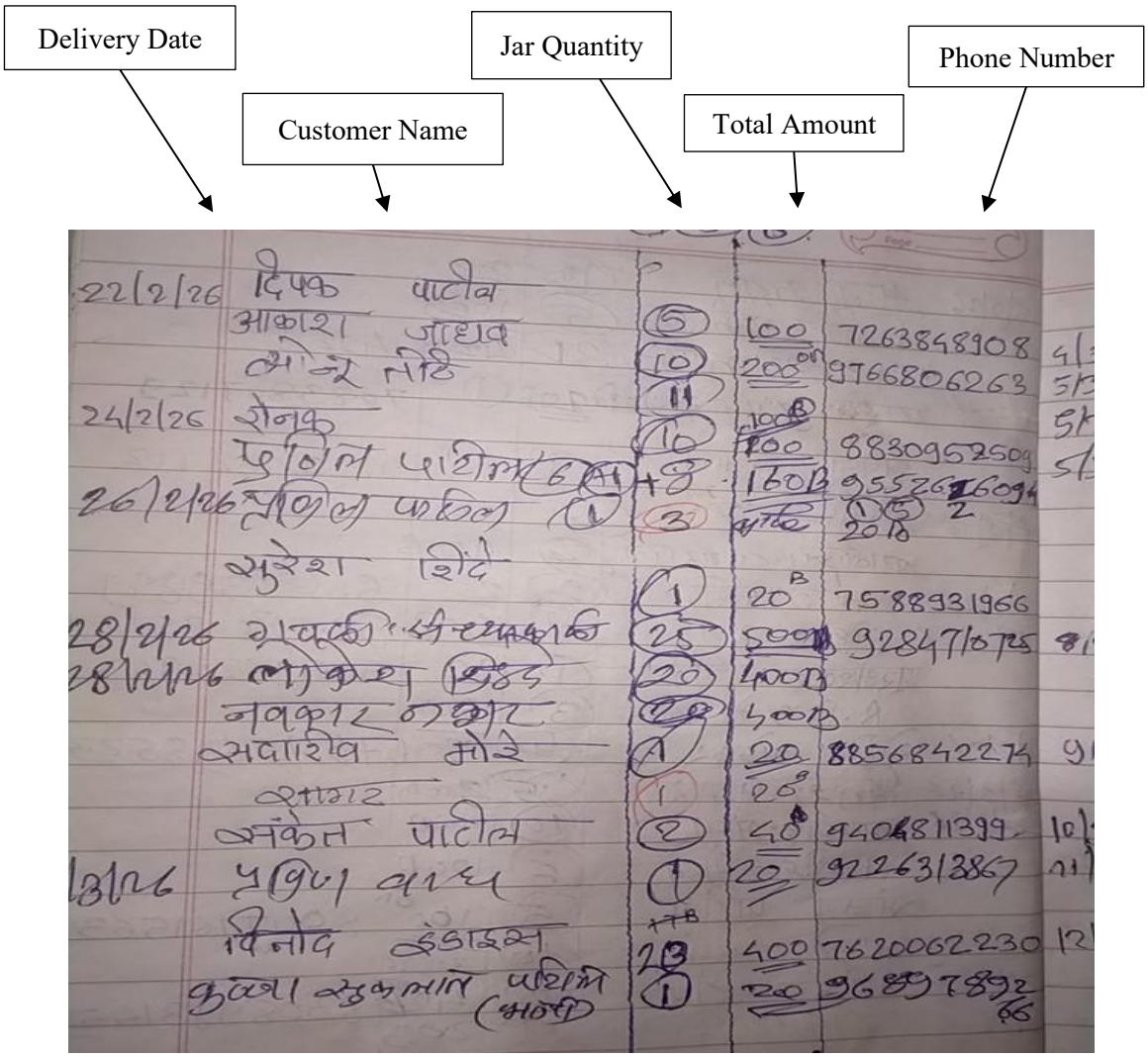


Fig.2.1 Manual Booking Record

- The supplier maintains order records manually in a notebook.
- The notebook contains order date, customer name, and number of water jars ordered.
- The supplier also writes the total payment amount based on the number of jars ordered.
- Each customer's phone number is recorded for contact and delivery purposes.
- All calculations and records are written by hand, which can make tracking and managing data difficult.

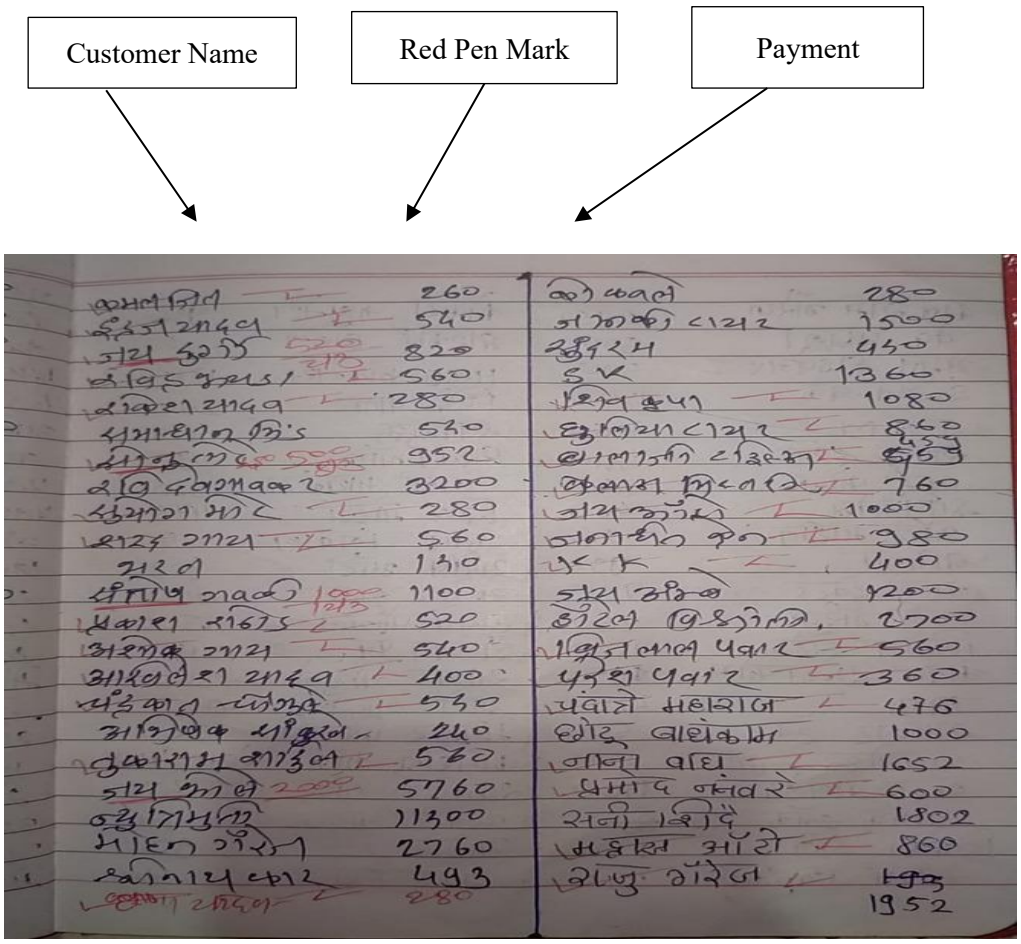


Fig.2.2 Manual Booking Record

- The supplier records regular customer names and total payment in a notebook.
- It shows the weekly or monthly amount of each customer.
- A red tick mark indicates payment is completed.
- All records are maintained manually.

2.3 LIMITATIONS OF EXISTING SYSTEMS

The existing water jar distribution system has several problems and limitations because it is based on manual work. The supplier manages all records using notebooks, which makes the process slow and difficult. It is hard to track orders, deliveries, and payment details properly. Sometimes, records may be missed or written incorrectly, which leads to confusion. Finding old records also takes a lot of time. There is no proper system to track pending orders and payments, which affects daily work. Managing a large number of customers becomes difficult, and the chances of errors increase. Because of these issues, the existing system is not efficient and needs improvement through a digital solution.

2.4 PROBLEM IDENTIFICATION

From studying the current system, many problems are found in the water jar distribution process. The main problem is that all work is done manually. It is difficult to manage orders, deliveries, and payments properly. Sometimes there is confusion in tracking which orders are pending and which payments are not completed.

It also takes a lot of time to find old records. As the number of customers increases, the work becomes more difficult and confusing.

So, there is a need for a better system to solve these problems and make the work easier.

2.5 NEEDS OF SYSTEM

The current system is manual, so it is difficult to manage all work properly. There is a need for a system that can make the process easy and organized. The system should help in managing orders, deliveries, and payment records in one place. It should reduce manual work and save time. It should also help in tracking pending orders and payments easily. There is also a need for proper record management so that data is not lost and can be accessed anytime. The system should improve accuracy and reduce mistakes. Overall, a digital system is needed to make the work faster, easier, and more efficient.

2.6 PROBLEM DEFINITION

The system is a web-based application design for managing water jar orders, deliveries, and payments in an easy way. In the current system, the water jar supplier handles customer orders and records manually using notebooks or phone calls. This makes it hard to keep track of daily orders, pending deliveries, and payment details. As business grows, mistakes and confusion increase, and work becomes slow. To solve this problem, a simple online system is needed that helps the supplier manage water jar supply work in a clear and organized manner.

CHAPTER 3
SCOPE OF THE PROJECT

3.1 SCOPE OF PROJECT

The Web Application for Water Jar Distribution System helps the supplier to manage water jar orders, deliveries, and payment records in an easy and organized way. This application is specially designed for a single supplier to handle all daily activities efficiently without depending on manual records.

The system allows the supplier to manage customer details, track orders placed by customers, and monitor delivery status in a proper way. It also helps in maintaining payment records, including both paid and pending payments. This reduces manual work, avoids confusion, and makes the process faster and more accurate.

The application enables the supplier to keep track of daily, weekly, and monthly orders in a structured format. All records are stored digitally in the database, so data can be accessed anytime when required. This makes it easier to find information quickly and helps in better management of the business.

In addition, the system improves communication between supplier, delivery staff, and customers. Delivery staff can view assigned orders and update delivery status, which helps in proper tracking of deliveries. Customers can also place orders and check their order status easily.

Overall, the system is designed to improve efficiency by reducing errors, saving time, and simplifying all operations. It provides a simple and user-friendly interface so that the supplier can easily use the system without any technical difficulty, making the entire water jar distribution process smooth and effective.

Order Management:

- Customers can place orders for water jars.
- The system stores all order details like quantity and date.
- Supplier can view and manage all orders in one place.

1. **Customer Management:**
   - Stores customer information like name and contact details.
   - Helps in managing regular and new customers easily.
2. **Delivery Management:**
   - Supplier can track delivery status (Delivered).
   - Helps in managing and assigning deliveries properly.

3. **Payment Tracking:**
   - Shows paid and pending payments clearly.
4. **Record Management:**
   - Maintains daily, weekly, and monthly records.
   - Stores data digitally for easy access and safety.
5. **Reporting:**
   - Provides simple reports of orders and payments.
   - Helps the supplier understand business activities better.

CHAPTER 4
METHODOLOGY

This chapter explains the methodology used to design and develop the Web Application for Water Jar Distribution System. It describes the step-by-step process followed during development, including requirement analysis, system design, implementation, and testing.

The project follows the Waterfall Model of Software Development Life Cycle (SDLC) to ensure a proper and organized development process. Each phase is completed carefully before moving to the next phase.

The purpose of this chapter is to provide a clear understanding of how the system is planned, designed, and implemented step by step.

4.1 SDLC APPROACH FOLLOWED WATERFALL MODEL

The Waterfall Model is selected because the project requirements are clear and can be implemented phase by phase. Each phase produces verified outputs before the next phase begins, ensuring systematic progress and better control.

Phases followed:

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Deployment
6. Maintenance

The following diagram shows the sequential phases of the Waterfall Model adopted for this project:

Figure 4.1: SDLC Waterfall Model — Sequential Phase Flow

4.2 REQUIREMENT ANALYSIS

Requirement analysis was performed through observation and discussion with the water jar supplier to understand the actual needs of the system.

- Discussion with the supplier to understand daily work and problems
- Observation of manual order, delivery, and payment process
- Study of common issues in the manual system like errors and confusion
- Identifying required features such as order management, delivery tracking, and payment management

Key Problems Identified in Existing / Manual Workflow

- Orders are not managed properly due to manual record keeping.
- Delivery status is not updated in real time.
- Confusion occurs in tracking pending and completed orders.
- Payment tracking is weak and not well organized.
- Records are time-consuming to check and may contain errors.

4.2.1 Admin Requirements

The supplier (admin) plays an important role in managing and controlling the overall system. Admin requirements focus on handling daily operations, maintaining accurate records, and monitoring all activities in the water jar distribution system.

- Supplier should be able to view all customer orders.
- Maintain categories and products
- Supplier should manage customer details.
- Supplier should update delivery status.
- Supplier should manage payment records.
- Supplier should view daily, weekly, and monthly order details

4.2.2 Delivery Staff Requirements

Delivery staff are responsible for handling the day-to-day delivery operations in the water jar distribution system. Their requirements focus on efficient execution of delivery tasks and updating delivery status.

- Delivery staff should be able to view only the orders assigned to them.
- Delivery staff should see customer name, address, and jar quantity.
- Delivery staff should update delivery status after delivery.

4.2.3 Customer Requirements

Customers interact with the system to place orders and manage their water

jar requirements. Their requirements focus on ease of use, clear information, and quick access to order and payment details.

- Customer should be able to place water jar orders online.
- Customer should enter name, contact number, address, and jar quantity.
- Customer should be able to view order details.
- Customer should get order confirmation.

4.3 SYSTEM ARCHITECTURE

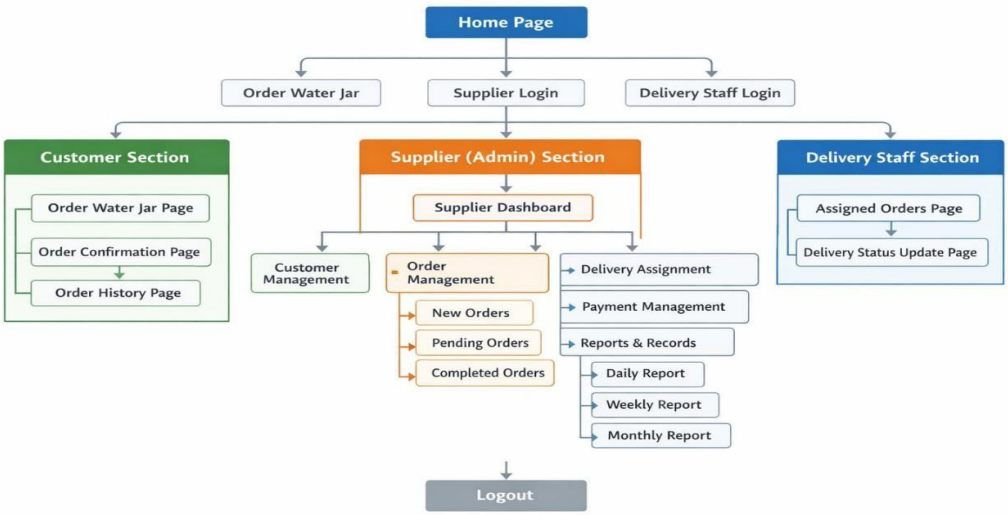


Figure 4.2: System Architecture

1. **Home Page**
   - The Home Page is the main entry point of the website.
   - From this page, users can choose Customer, Supplier, or Delivery Staff options.
2. **Customer Section**
   - Customers place water jar orders through the Order Water Jar page.
   - After placing an order, customers receive order confirmation.
3. **Supplier (Admin) Section**
   - The supplier logs in to the system using the Supplier Login page.
   - After login, the supplier accesses the Supplier Dashboard.
   - The supplier manages customer details and views all orders.
   - Orders are managed as new, pending, and completed orders.
   - The supplier assigns orders to delivery staff.
   - The supplier manages payment status and views reports.
   - Daily, weekly, and monthly records are available for analysis.

4. **Delivery Staff Section**
    - Delivery staff logs in using the Delivery Staff Login page.
    - Delivery staff view only the orders assigned to them
    - After delivery, staff updates the delivery status in the system.
5. **Logout**
    - Supplier and delivery staff can safely exit the system using the logout option

4.4 ACTION PLAN

- Finalization of project team and allocation of project guide
- Identification of project domain (Water Jar Distribution System)
- Submission of project idea by the team
- Approval of project idea by project guide and HOD
- Study of existing system and problem identification
- Requirement gathering from water jar supplier
- Preparation of project planning and methodology
- Preparation and submission of project planning report
- Design of user interface (UI) for web application
- Design of database structure and creation of database (MongoDB)
- Development of frontend using HTML, CSS, Bootstrap, and React.js
- Development of backend using Node.js and Express.js
- Implementation of modules like customer management, order management, delivery management, and reports
- Integration of frontend with backend (API development)
- Database connectivity and data handling
- Testing of the system and fixing errors
- Making changes based on suggestions from guide
- Preparation of project presentation and documentation
- Final testing and verification of system
- Submission of final project report and project work

4.5 TECHNOLOGY USE TO DEVELOP PROJECT

- **Programming Language:** JavaScript
- **Frontend:** HTML, CSS, Bootstrap, React.js
- **Backend:** Node.js, Express.js

- **Database:** MongoDB
- **Development Tools:** GitHub (Version Control), Postman (API Testing), Visual Studio Code (Development)
- **Operating System:** Windows 11

This chapter describes the development methodology of the Water Jar Distribution System, including requirement analysis, system design, and implementation approach. It explains how different modules such as customer management, order management, delivery management, and report generation are designed and integrated into the system. The methodology ensures that the system is reliable, easy to use, and efficient for managing water jar distribution operations.

CHAPTER 5

DETAILS OF DESIGNS, WORKING AND PROCESSES

This chapter provides a detailed explanation of the system design, workflow, and working process of the Web Application for Water Jar Distribution System. It explains how different parts of the system work together and how data moves within the system. The chapter includes system flow, module-wise working, database structure, and the complete working process from order placement to delivery and payment tracking.

The purpose of this chapter is to give a complete understanding of how the system functions internally.

5.1 SYSTEM FLOW DIAGRAM

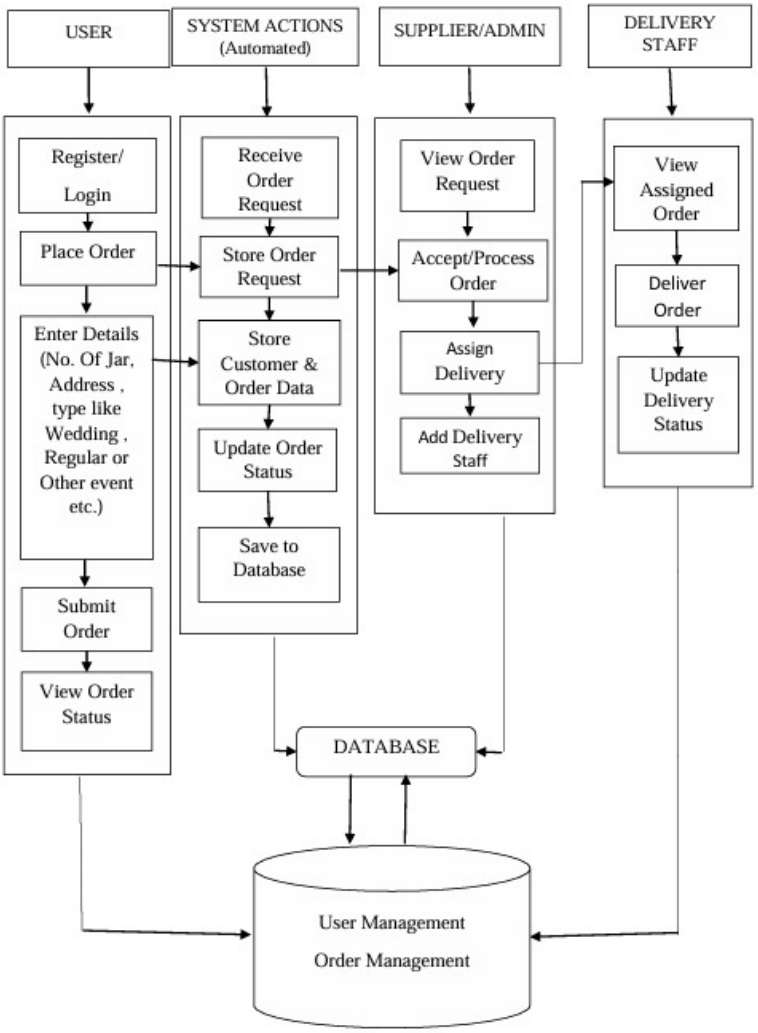


Figure 5.1: System Flow Diagram

This diagram represents the complete working process of the Aqua Jar Delivery System, starting from user order placement to final delivery and status update. The system

is divided into four main parts: User, System Actions, Supplier/Admin, and Delivery Staff, all connected through a central database.

1. **User (Customer Module)**

The process starts with the user.

First, the user registers or logs into the system. After successful login, the user places a new order.

The user enters all required details such as:

- Number of water jars
- Delivery address
- Order type (Regular, Wedding, or other events)

After filling in the details, the user submits the order. Once the order is submitted, the user can also view the order status anytime to track progress

2. **System Actions (Automated Process)**

After the user submits the order, the system automatically performs several actions.

- It receives the order request from the user
- Stores the order details safely
- Saves customer and order data
- Updates the order status based on progress
- Finally, all the information is saved into the database

This part of the system works automatically without human intervention, ensuring smooth and fast processing.

3. **Supplier / Admin Module**

Once the order is stored, the supplier or admin can access it.

- Supplier views all order requests
- Then accepts or processes the order
- After that, the supplier assigns the delivery to a delivery staff member
- Supplier can also add or manage delivery staff

This module is responsible for managing orders and ensuring they are properly assigned for delivery.

4. **Delivery Staff Module**

After assignment, the delivery staff takes action.

- Delivery staff views assigned orders
- Delivers the water jars to the customer
- After delivery, they update the delivery status in the system

This ensures that the system always has updated information about the order.

5. Database

The database plays an important role in the system. It stores all data such as User details, Order details and Delivery status. All modules (User, System, Supplier, Delivery Staff) are connected to the database, making data access and updates easy and efficient.

5.2 CUSTOMER MODULE

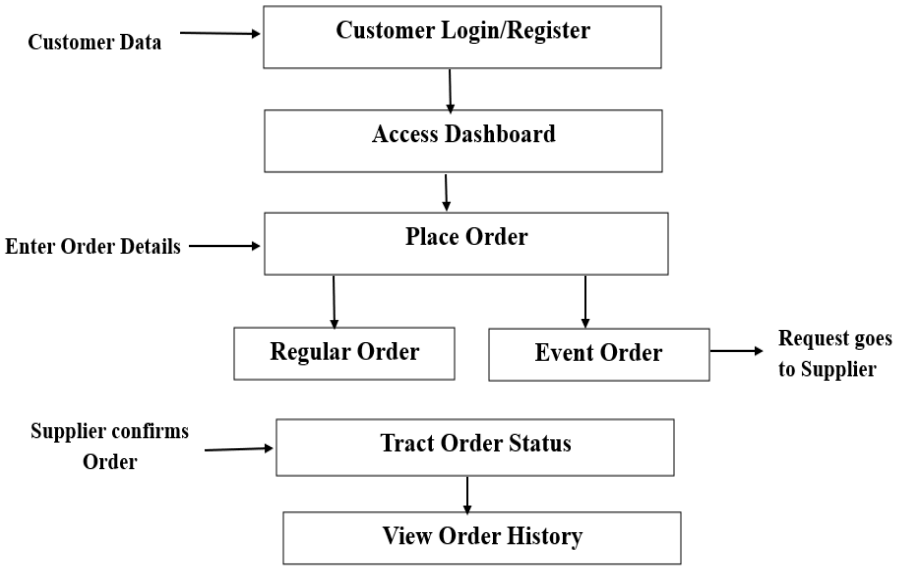


Figure 5.2: Customer Module Flow Diagram

This diagram explains the complete flow of how a customer interacts with the Aqua Jar Delivery System. It shows each step starting from customer login to placing an order, selecting order type, and tracking order status. The process is designed to be simple, user-friendly, and efficient.

1. Customer Data & Login/Register

The process begins with customer data input.

The customer either registers as a new user or logs into the system.

- If the customer is new, they must provide details like name, email, phone number, and password to create an account.
- If the customer already has an account, they can log in using their credentials.

This step ensures that only authorized users can access the system and place orders.

2. Access Dashboard

After successful login or registration, the customer is directed to the dashboard.

- The dashboard acts as the main interface of the system.
- It provides options such as placing a new order, viewing existing orders, and checking order status.
- It may also display summary information like total orders, pending orders, or delivered orders.

This makes it easy for the customer to navigate and use the system.

3. Enter Order Details & Place Order

From the dashboard, the customer selects the Place Order option.

The customer needs to enter important details such as:

- Number of water jars required
- Delivery address
- Type of order (regular or event-based)

The system collects and validates this information before proceeding.

Once all details are entered correctly, the customer proceeds to place the order.

4. Order Type Selection (Regular / Event Order)

After entering order details, the customer selects the type of order:

Regular Order

- Used for daily or routine water delivery
- Usually for home or office use

Event Order

- Used for special occasions like weddings, parties, or other events
- May involve larger quantity of jars

After selecting the type, the order request is sent to the supplier/admin for further processing.

5. Request Goes to Supplier

Once the order is placed:

- The system forwards the order request to the supplier
- Supplier can view all incoming orders
- The request contains all details entered by the customer

This step connects the customer module with the supplier module.

6. Supplier Confirms Order

The supplier reviews the order and takes action:

- Checks order details and availability
- Accepts or confirms the order

- Updates the order status in the system

If required, the supplier may assign delivery staff for the order.

7. Track Order Status

After placing the order, the customer can track the order status.

The system updates status in real-time

Common statuses include:

- Pending (order placed but not confirmed)
- Confirmed (order accepted by supplier)
- Delivered (order completed)

This feature provides transparency and keeps the customer informed.

8. View Order History

- Finally, the customer can access the order history section.
- Displays all previous and current orders
- Includes details such as Order type, Quantity of jars, Status, Date
- This helps customers to review past orders and manage their usage.

5.3 SUPPLIER MODULE

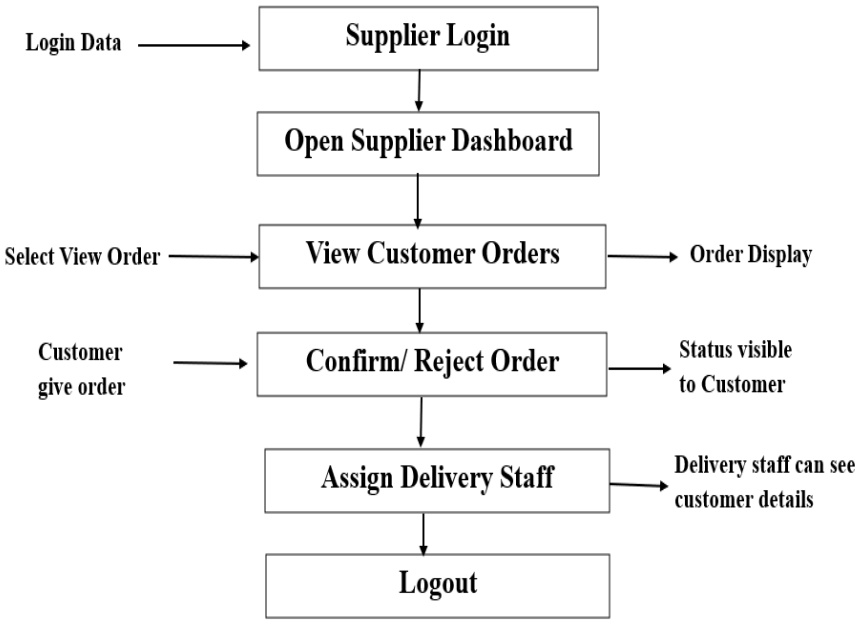


Figure 5.3: Supplier Module Flow Diagram

This diagram explains the working process of the Supplier (Admin) module in the Aqua Jar Delivery System. It shows how the supplier manages orders from login to assigning delivery staff.

1. **Supplier Login**

The supplier logs into the system by entering email and password. The system checks the login details and allows access only if they are correct. This step is important for security so that only authorized users can use the system.

2. **Open Supplier Dashboard**

After login, the supplier is taken to the dashboard. The dashboard is the main page where the supplier can see an overview of system activities. It helps the supplier to easily navigate to different sections like orders, customers, and assignments.

3. **View Customer Orders**

The supplier can see all orders placed by customers. Each order is displayed with details such as customer name, order type, number of jars, and delivery date. This helps the supplier understand all incoming orders clearly.

4. **Confirm / Reject Order**

After viewing the orders, the supplier takes action. The supplier can either confirm the order if it can be completed or reject it if there is any issue. Once the action is taken, the system updates the order status.

The updated status becomes visible to the customer, so they know whether their order is accepted or not.

5. **Assign Delivery Staff**

If the order is confirmed, the supplier assigns it to delivery staff. The supplier selects a delivery person based on availability or workload. After assignment, the delivery staff can see the order and customer details.

This step ensures that the order is delivered properly.

6. **Logout**

After completing all tasks, the supplier can log out of the system.

5.4 DELIVERY STAFF MODULE

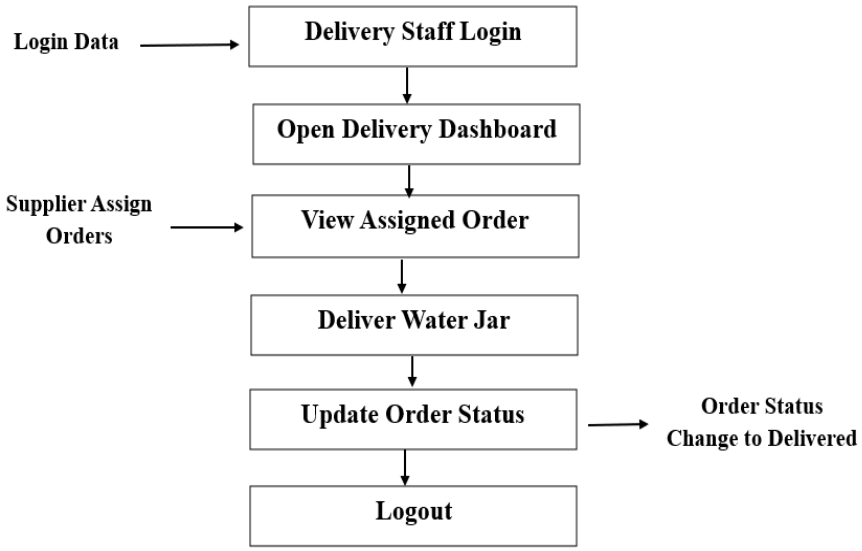


Figure 5.4: Delivery Staff Module Flow Diagram

This diagram explains how the delivery staff works in the System. It shows the complete process from login to delivering the order and updating its status.

1. **Delivery Staff Login**

   The process starts when the delivery staff enters login data such as email and password. The system verifies these details and allows access only if they are correct. This ensures that only authorized delivery staff can use the system.

2. **Open Delivery Dashboard**

   After successful login, the delivery staff is directed to the delivery dashboard. The dashboard is the main page where the delivery staff can see an overview of their assigned tasks and delivery activities.

3. **View Assigned Order**

   The supplier assigns orders to delivery staff.

   In this section, the delivery staff can view all assigned orders.

   Each order shows details such as:

   - Customer name
   - Delivery address
   - Number of jars
   - Delivery date

   This helps the delivery staff understand what needs to be delivered.

4. **Deliver Water Jar**

   After checking the assigned orders, the delivery staff proceeds to deliver the water

jars to the customer.

- The delivery staff visits the customer location
- Delivers the required number of jars

This step completes the physical delivery process.

5. Update Order Status

After successful delivery, the delivery staff updates the order status in the system.

- The status is changed to "Delivered"
- This update is saved in the system

This step is important because it informs both the supplier and the customer that the order has been completed.

6. Logout

After completing all delivery tasks, the delivery staff can log out of the system.

5.5 USE CASE DIAGRAM

The Use Case Diagram identifies the different users (actors) of the Water Jar Distribution System and shows the actions they can perform within the system. It provides a high-level view of how users like customers, supplier, and delivery staff interact with the system and its features.

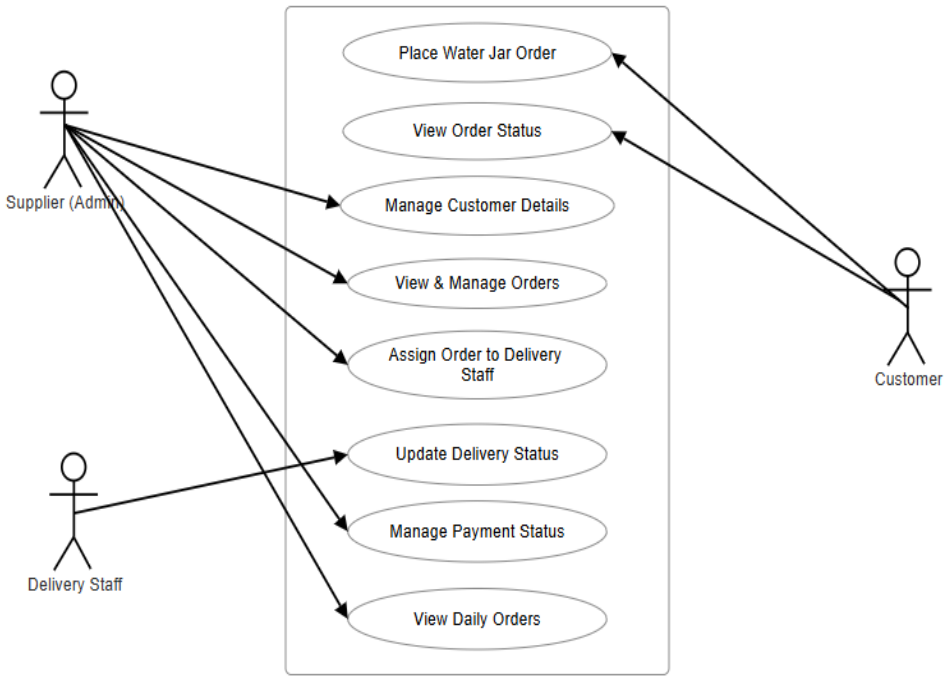


Figure 5.5: Use Case Diagram — System Actors and Role-wise Use Cases

Actors and Their Functions

1. **Customer**
   - Customer can place water jar order by entering required details such as number of jars, delivery address, and order type (regular or event).
   - Customer can view order status to check whether the order is pending, confirmed, or delivered.
   - Customer can track order progress, which helps in knowing the current stage of the order.
   - This actor mainly interacts with the system for ordering and monitoring deliveries.
2. **Supplier (Admin)**
   - Supplier can manage customer details, including viewing and updating customer information.
   - Supplier can view and manage all orders placed by customers, which helps in understanding order requirements.
   - Supplier can confirm or process orders after reviewing the details.
   - Supplier can assign orders to delivery staff based on availability and workload.
   - Supplier can manage payment status, such as checking advance and remaining payment.
   - Supplier can view daily orders to monitor overall system activity.
   - This actor plays the main role in controlling and managing the system operations.
3. **Delivery Staff**
   - Delivery staff can view assigned orders given by the supplier.
   - Delivery staff can deliver water jars to the customer location.
   - After delivery, they update the delivery status in the system (e.g., delivered).
   - This ensures that the system has accurate and updated information.
4. **System Functions**
   - The system allows order placement and tracking by customers.
   - It supports order management by the supplier.
   - It provides delivery assignment and tracking features.
   - It maintains payment status management for each order.
   - It ensures real-time status updates for all users.

CHAPTER 6

RESULT AND APPLICATION

This chapter presents the results of the developed Water Jar Distribution System through its main web pages and their functions. The screenshots show how different users interact with the system for activities such as placing orders, managing customer details, handling delivery, updating order status, and generating reports.

6.1 HOME PAGE

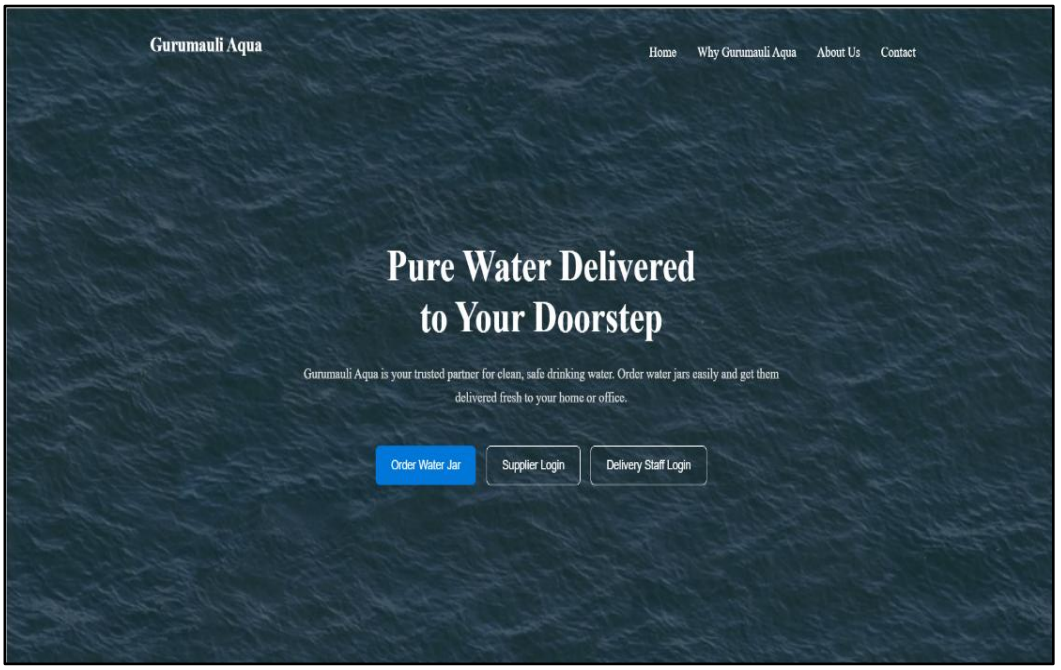


Figure 6.1: Home Page

The Home Page is the main entry point of the Water Jar Distribution System. It displays the name "Gurumauli Aqua" along with a navigation menu for easy access to different sections of the website such as Home, About Us, and Contact. A large heading highlights the main purpose of the system, which is to provide pure water delivery to customers' doorsteps. The page also includes a short description of the service.

This page contains important buttons such as:

- Order Water Jar button – allows customers to place an order
- Supplier Login button – used by admin/supplier to log in
- Delivery Staff Login button – used by delivery staff to access their dashboard

6.2 REGISTRATION FORM FOR NEW CUSTOMER

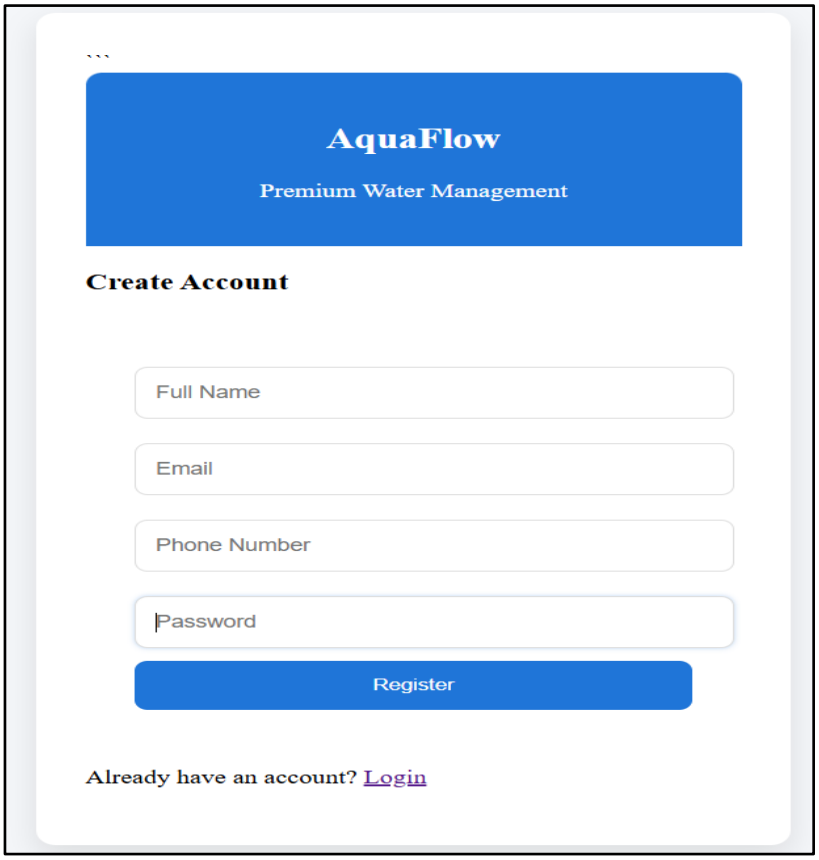


Figure 6.2: Registration Page

This page allows new users to create an account in the system. It includes input fields such as full name, email, phone number, and password. Users can register using the register button. There is also a login link for users who already have an account.

Components Used:

- Input fields (Full Name, Email, Phone Number, Password)
- Register button
- Login link
- Card layout

```
const handleSubmit = async (e) => {
e.preventDefault();
const validationError = validate();
if (validationError) {
  setError(validationError);
```

```
    return;
  }
  try {
    await API.post("/auth/register", form);
    navigate("/");
  } catch (err) {
    setError(err.response?.data?.message || "Registration failed");
  }
  };
```

Table 6.1: Registration Validations – Secure Credential-Based Authentication

Fields	Validation Applied
Full Name	"^[A-Za-z]{3,30}$", "Name must contain only alphabets and spaces (3–30 characters)"
Email	"^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+.[A-Za-z]{2,}$", "User must enter valid email format
Password	"(?=.\d)(?=.[a-z])(?=.*[A-Z]).{8,16}", "Password must contain at least one number, one uppercase letter, one lowercase letter, and length between 8 to 16 characters"
Empty Fields Validation	"All fields (Name, Email, Phone, Password) must not be empty"
Invalid Email	"Display error message if email format is incorrect"

6.3 CUSTOMER LOGIN PAGE

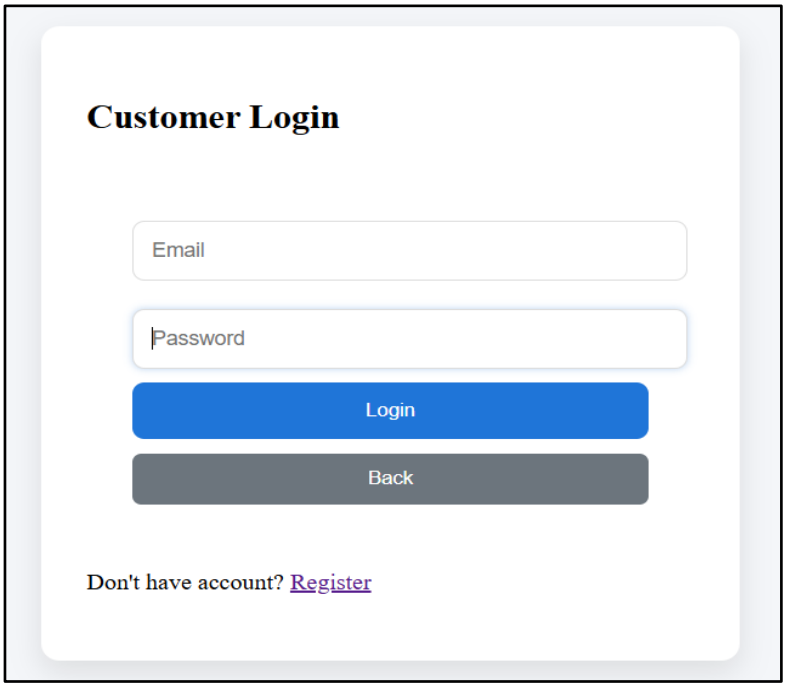


Figure 6.3: Customer Login Page

This page allows customers to log in to the system. It contains input fields for email and password. Users can click the login button to access their account or the back button to return to the previous page. There is also a register link for new users who do not have an account.

Components Used:

- Input fields (Email, Password)
- Buttons (Login, Back)
- Register link
- Card layout

```
const handleSubmit=async(e)=>{
  e.preventDefault();
  try{
    const res = await API.post("/auth/login",form);
    localStorage.setItem("token",res.data.token);
    localStorage.setItem("role",res.data.user.role);
    localStorage.setItem("name",res.data.user.name);
    navigate("/customer/dashboard");
  }catch(err){
    setError(err.response?.data?.message || "Login failed");
```

6.4 SUPPLIER LOGIN PAGE

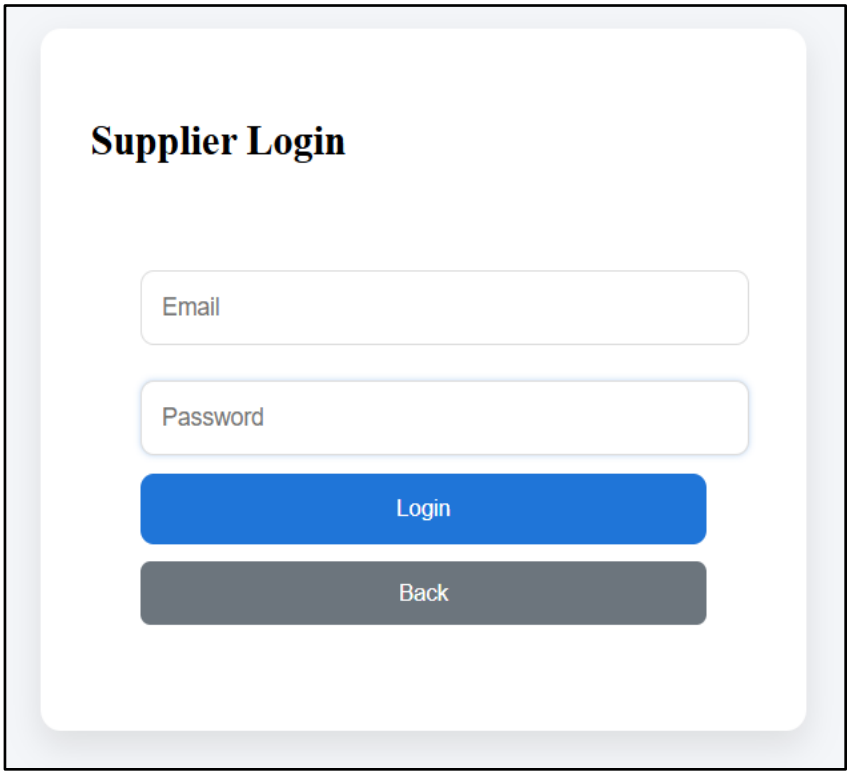


Figure 6.4: Supplier Login Page

This page is designed for suppliers to log in to the system and access their dashboard. It provides a simple and user-friendly interface where the supplier can enter login details such as email and password. The main purpose of this page is to allow only authorized users (suppliers) to access and manage the system.

The page includes input fields for entering the email and password. After entering the correct details, the supplier can click on the login button to access the supplier dashboard, where they can manage orders, customers, deliveries, and payments. If the supplier wants to go back to the previous page, a back button is provided for easy navigation.

The layout of the page is clean, simple, and centered, making it easy to use without any confusion. Basic validation can also be applied to ensure that correct details are entered before login.

Components Used:

- Input fields (Email, Password)
- Buttons (Login, Back)
- Centered card layout

6.5 DELIVERY STAFF LOGIN PAGE

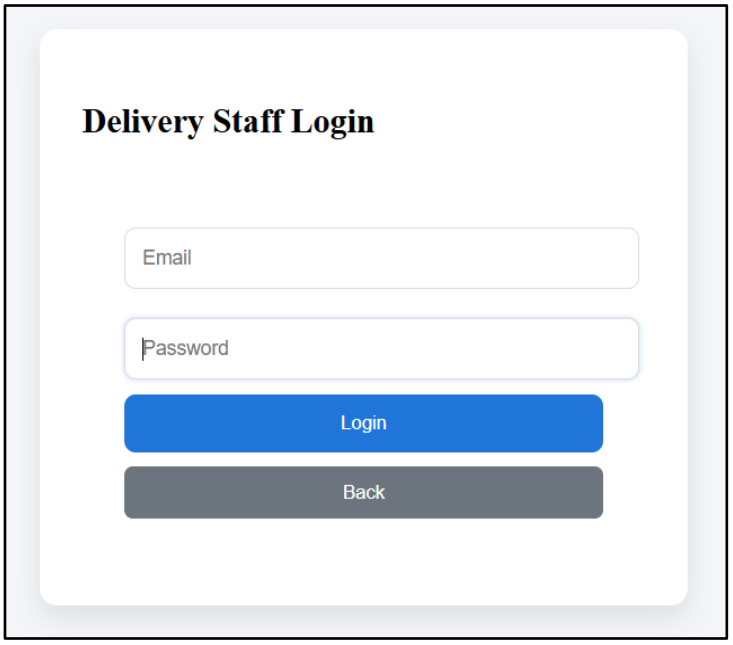


Figure 6.5: Delivery Staff Login Page

This page is used by delivery staff to log in to the system. It contains input fields for email and password. Users can log in using the login button or return using the back button. The design is simple and user-friendly.

- Components Used:
- Input fields (Email, Password)
- Buttons (Login, Back)
- Card layout

6.6 CUSTOMER DASHBOARD

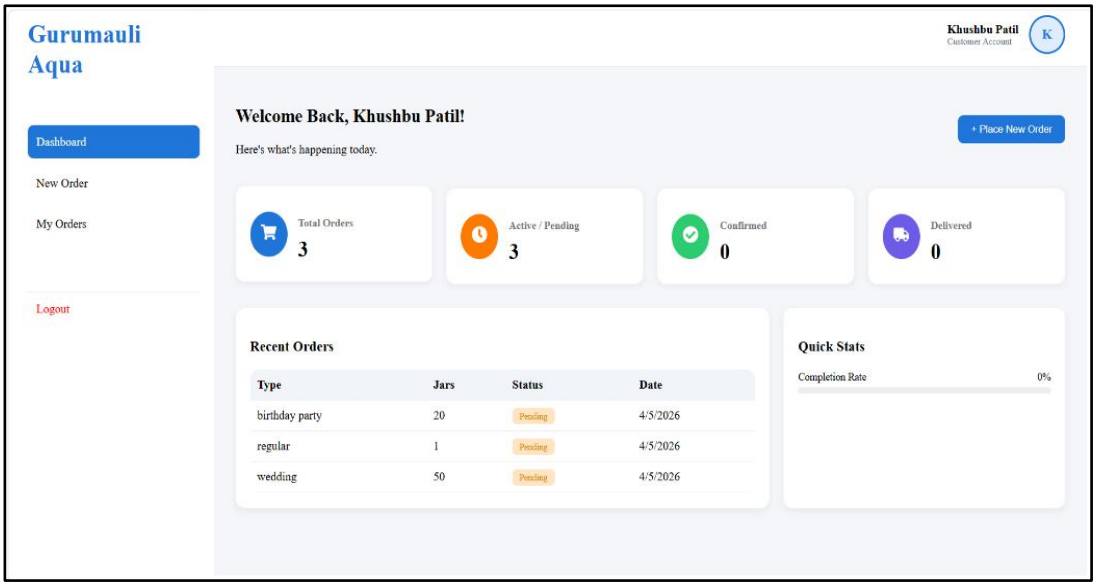


Figure 6.6: Customer Dashboard

Customer dashboard contains header, profile section and menu at left side. The menu includes options like Dashboard, New Order, My Orders and Logout. The dashboard shows overall information about customer orders.

The top section displays welcome message with customer name and a button to place new order.

The dashboard shows summary cards which represent:

- Total number of orders placed by customer
- Active / Pending orders
- Confirmed orders
- Delivered orders

The table shows the details of recent orders such as order type, number of jars, order status and date.

Customer can:

- Place new order using “Place New Order” button
- View all previous orders
- Track current order status

6.7 CUSTOMER ORDER PAGE

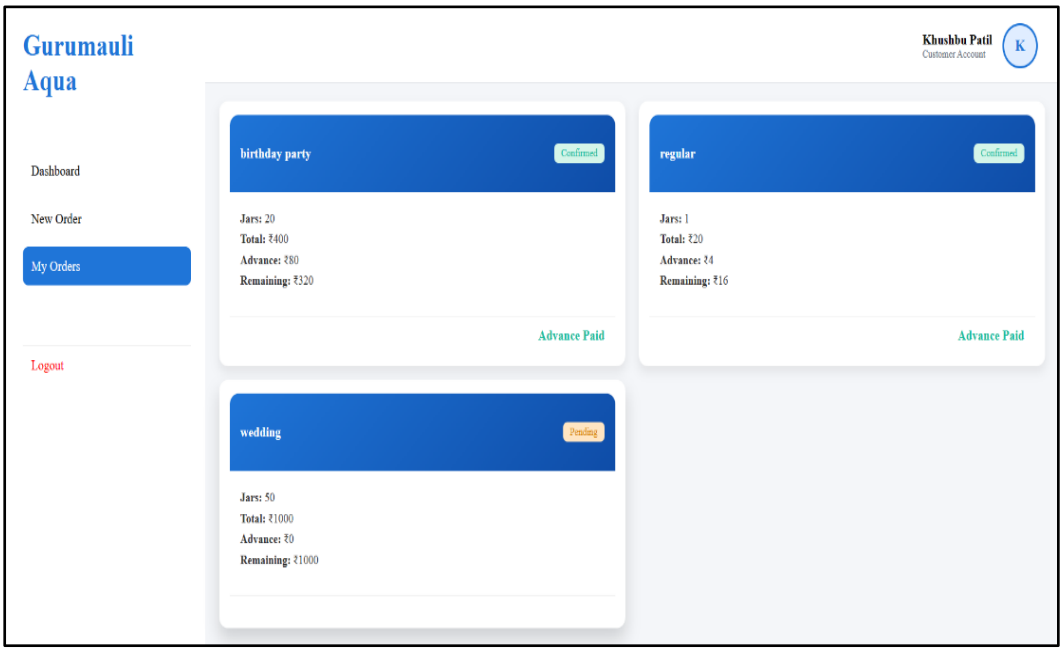


Figure 6.7: Customer Order Page

Customer My Orders page contains header, profile section and menu at left side. The menu includes options like Dashboard, New Order, My Orders and Logout. This page shows all the orders placed by the customer in card format. Each order is displayed

in a separate card. The card shows details such as:

- Order type (birthday, regular, wedding, etc.)
- Number of jars
- Total amount
- Advance paid
- Remaining amount

Customer can:

- View complete details of each order
- Check payment information (advance and remaining amount)
- Track order status

Status labels are used on each order card to show progress as follows:

1. Green – Confirmed (Order is accepted and processing)
2. Orange – Pending (Order is placed but not yet confirmed)

If advance payment is done, it is indicated as "Advance Paid" on the order card.

6.8 NEW ORDER PAGE

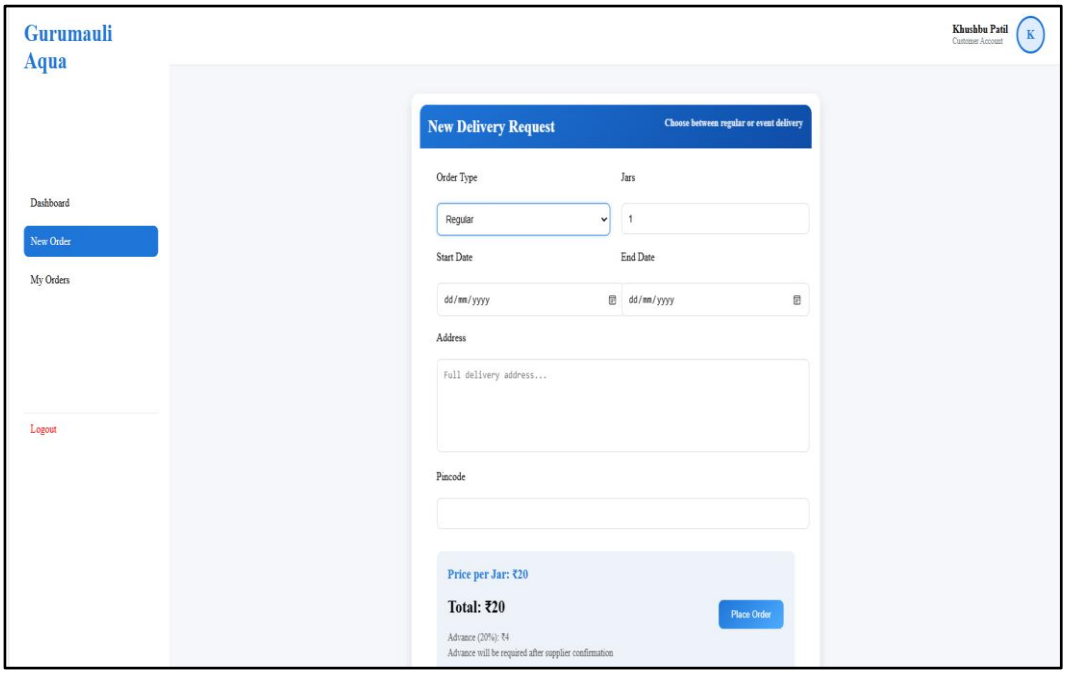


Figure 6.8: New Order Page

The page contains a form where customer can enter order details such as:

- Order type (Regular or Event delivery)
- Number of jars
- Start date and End date

- Delivery address
- Pincode

Customer can:

- Select order type based on requirement
- Enter delivery details and schedule
- Place new order using “Place Order” button

The page also shows pricing details such as price per jar and total amount, which is automatically calculated based on number of jars.

```
const handleSubmit = async (e) => {
  e.preventDefault();
  console.log("Submit clicked");
  if (!validateForm()) return;
  try {
    console.log("Calling API...");
    const res = await API.post("/orders/create", {
      ...form,
      totalAmount: calculateTotal(),
    });
    console.log("API RESPONSE:", res);
    alert("Order placed successfully!");
    navigate("/customer/dashboard");
  } catch (error) {
    console.error("FULL ERROR:", error.response || error);
    alert("Error placing order");
  }
};
  const total = calculateTotal();
  const advance = Math.floor(total * 0.2);
```

Table 6.2: New Order Page Validations

Field Name	**Validation / Description**
Order Type	"Required field" , "User must select order type (Regular/Event)"
Jars	"^[1-9][0-9]*$" , "Only positive numbers allowed (minimum 1 jar)"
Start Date	"Required field" , "Start date must be current or future date"
End Date	"Required field" , "End date must be greater than or equal to start date"
Address	"^[A-Za-z0-9 ,.-]{10,100}$" , "Enter valid delivery address (10–100 characters)"
Pincode	"^[0-9]{6}$" , "Only 6 digit numeric pincode allowed"
Price per Jar	"Auto-calculated field" , "Displayed based on system pricing"
Total Amount	"Auto-calculated field" , "Calculated based on number of jars"

6.9 SUPPLIER DASHBOARD PAGE

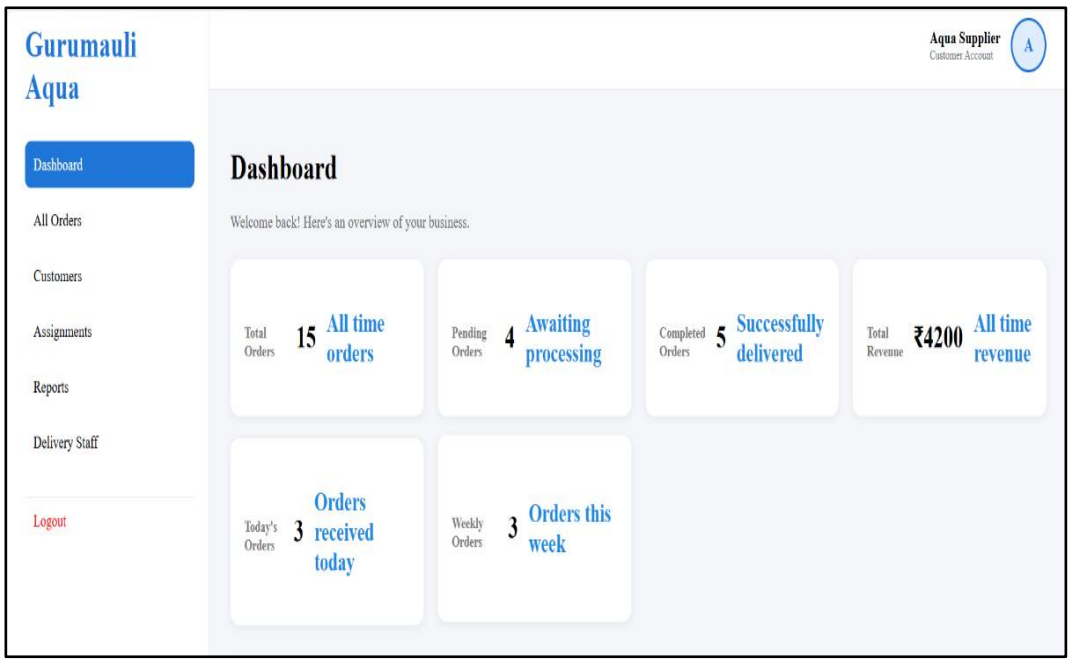


Figure 6.9: Supplier Dashboard

Supplier dashboard contains header, profile section and menu at left side. The menu includes options like Dashboard, All Orders, Customers, Assignments, Reports, Delivery Staff and Logout. This page provides an overview of supplier business activities.

The top section displays a welcome message for the supplier.

The dashboard shows summary cards which represent:

- Total number of orders (All time orders)
- Pending orders (Awaiting processing)
- Completed orders (Successfully delivered)
- Total revenue generated

Additional cards display:

- Today's orders received
- Weekly orders count

Supplier can:

- View all orders and manage them
- Check customer details
- Assign delivery tasks to delivery staff
- View reports and revenue details
- Manage delivery staff

Status indicators are used to show order progress as follows:

1. Pending – Awaiting processing
2. Completed – Successfully delivered

6.10 ALL ORDERS PAGE

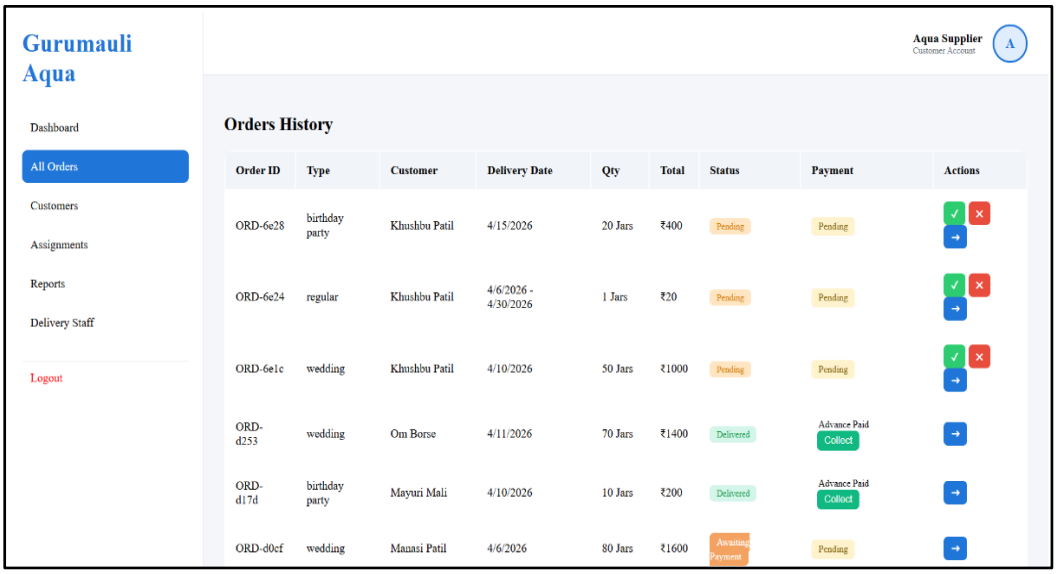


Figure 6.10: All Orders Page

This page displays all customer orders in a structured table format for easy tracking and management. It allows the supplier to monitor order details, status, payment information, and perform necessary actions on each order.

Supplier can:

- View complete list of all orders
- Check order details (type, customer, quantity, date, amount)
- Monitor order status (Pending, Delivered, etc.)
- Check payment status (Pending, Advance Paid, Collected)
- Accept or process new orders
- Reject or cancel orders if required
- Update order status (e.g., mark as delivered)
- Verify and update payment status
- Assign order to delivery staff (next step action button)
- Manage and track all orders efficiently

```
const fetchOrders = async () => {
  try {
    const res = await API.get("/orders/all");
    setOrders(res.data || []);
```

```
  } catch (error) {
    console.error("Error fetching orders:", error);
    alert("Failed to load orders");
  } finally {
    setLoading(false);
  }
};
```

6.11 ORDER ASSIGNMENT PAGE

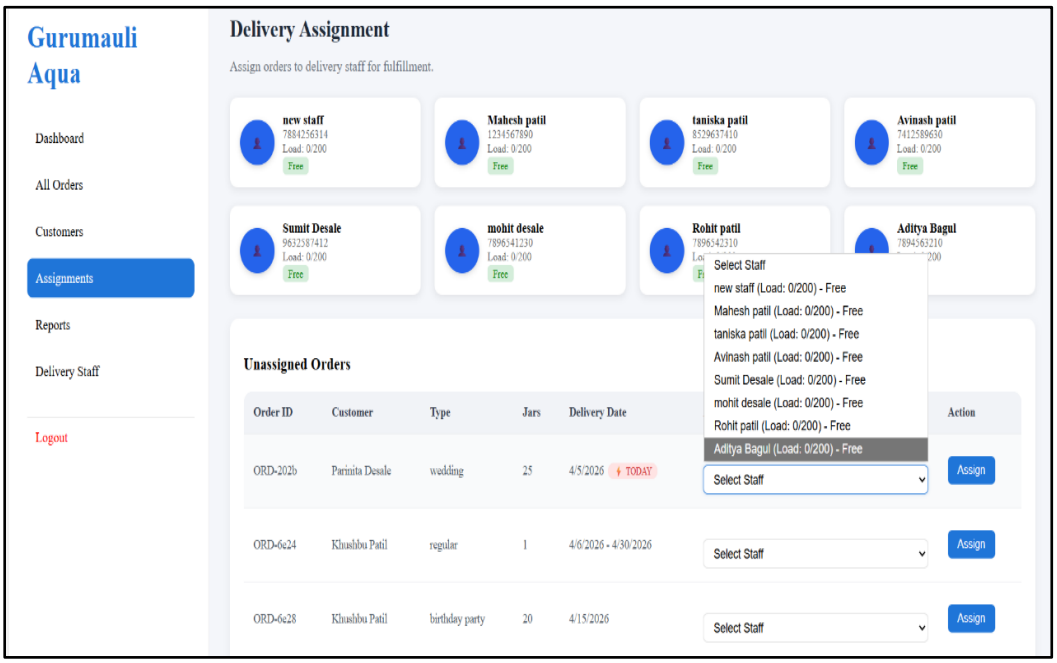


Fig. 6.11: Order Assignment Page

This page is used to assign customer orders to delivery staff for order fulfillment. It helps the supplier manage staff workload and ensure timely delivery of orders. At the top, delivery staff cards are displayed showing staff name and current load (number of assigned orders). This helps in selecting available staff.

Below that, a table of unassigned orders is shown.

The table includes:

- Order ID
- Customer name
- Order type
- Number of jars
- Delivery date

Supplier can:

- View all unassigned orders
- Select delivery staff from dropdown list
- Assign order to available staff using “Assign” button
- Check staff workload before assigning

6.12 ASSIGNED ORDER DETAILS

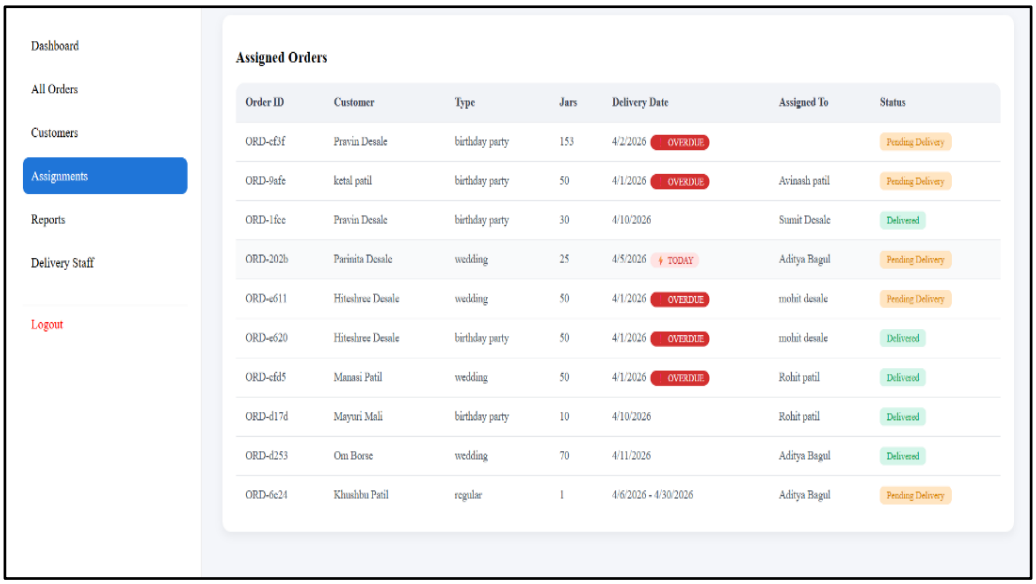


Figure 6.12: Assigned Order Details

This page shows all orders that are already assigned to delivery staff. It helps the supplier track delivery status and assigned staff for each order.

The table shows:

- Order ID
- Customer name
- Order type
- Number of jars
- Delivery date
- Assigned staff
- Order status

Supplier can:

- View which staff is assigned to each order
- Track delivery status (Pending Delivery, Delivered, Overdue)
- Monitor ongoing deliveries

Status labels are used as follows:

1. Pending Delivery – Order is assigned but not delivered

2. Delivered – Order completed
3. Overdue – Delivery date passed but not completed

```
const handleAssign = async (orderId) => {
 const staffId = selectedStaff[orderId];
 console.log("Assigning:", { orderId, staffId });
 if (!staffId) {
  return alert("Please select staff before assigning");
 }
 try {
  await API.put(`/assignments/assign/${orderId}`, {
   staffId,
  });
  fetchData();
 } catch (err) {
  console.log(err.response?.data || err.message);
  alert("Assignment failed");
 }
};
```

6.13 REPORT MANAGEMENT

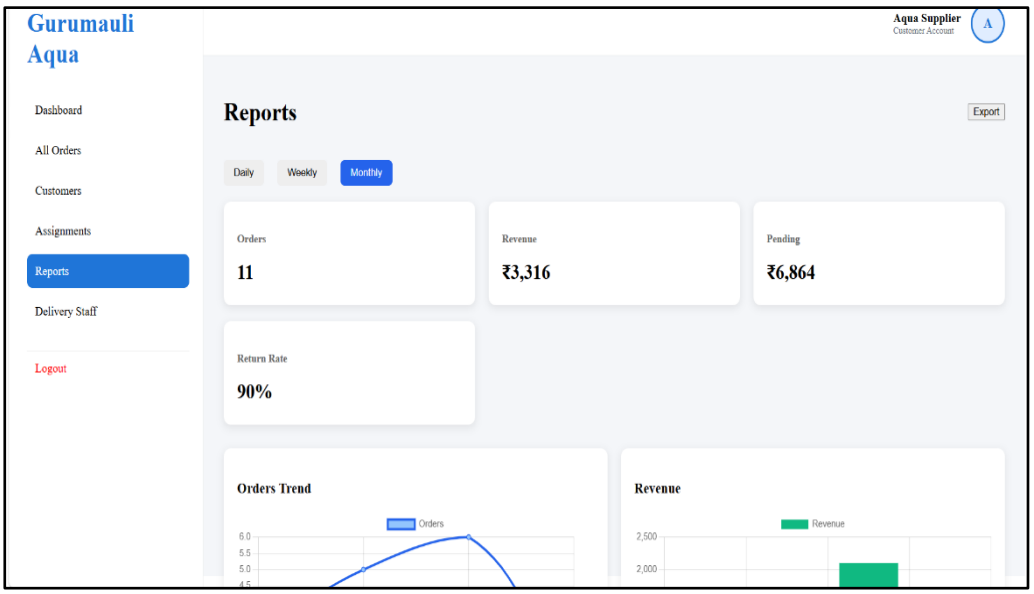


Figure 6.13: Reports Page I

This page shows analytical reports of orders and revenue for better business insights. It helps the supplier track performance using summary data and visual charts.

At the top, filter options are available such as:

- Daily
- Weekly
- Monthly

The page shows summary cards which include:

- Total number of orders
- Total revenue
- Pending amount
- Return rate

Below that, charts are used to visualize data:

- Orders Trend (line chart showing order activity over time)
- Revenue Chart (bar chart showing revenue comparison)
- Advance vs Remaining (pie chart showing payment distribution)

Supplier can:

- View business performance over time
- Analyze order trends and revenue
- Switch between daily, weekly and monthly reports
- Export report data

6.14 REPORT PAGE II

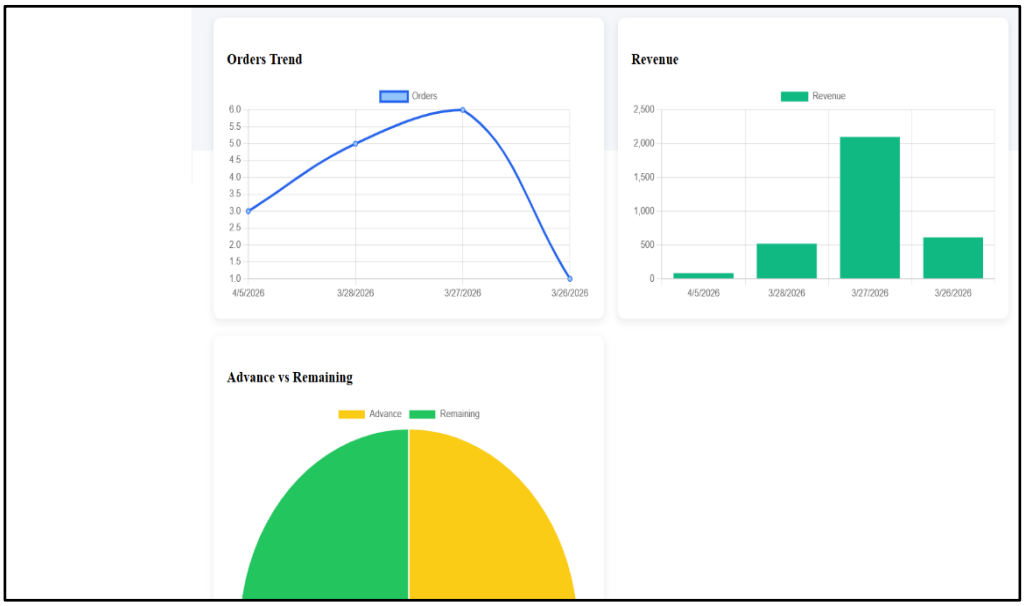


Figure 6.14: Report Page II

This image shows the dashboard of the Water Jar Distribution System. It helps the supplier or admin to quickly understand the business performance using Charts and

Graphs.

At the top, there are two charts:

- **Orders Trend:**

  This line graph shows how the number of orders changes over different dates. It helps to understand whether orders are increasing or decreasing over time.

- **Revenue:**

  This bar chart shows how much money is earned on different dates. It helps the supplier track daily or weekly income.

Below, there is another chart:

- **Advance vs Remaining:**

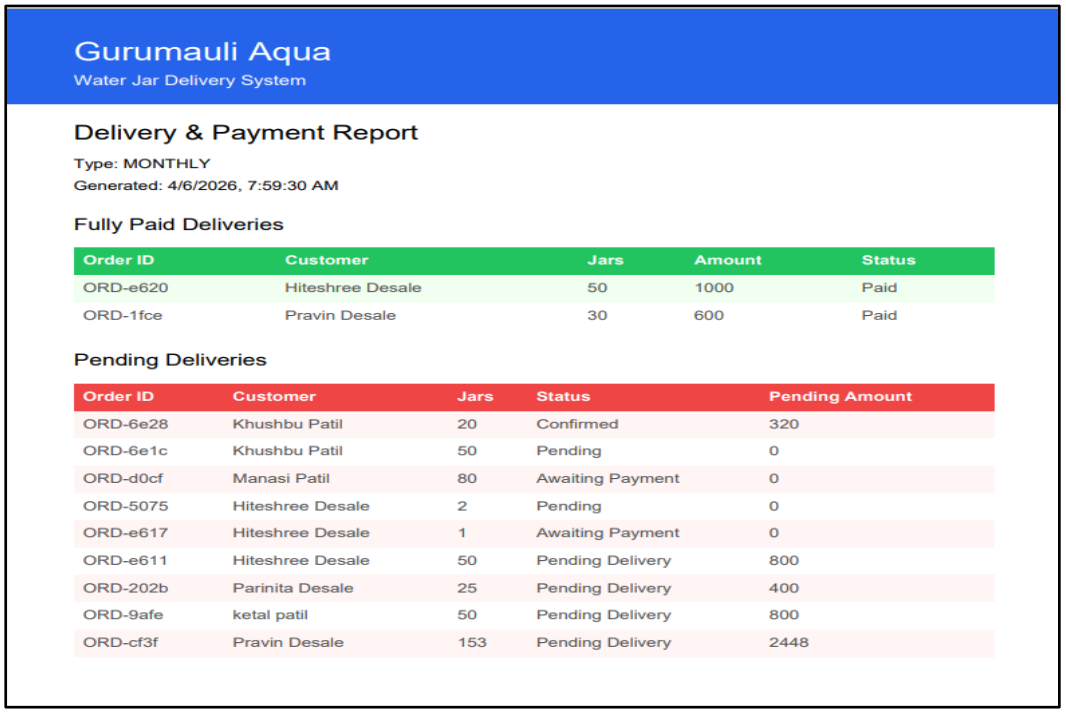
  This pie chart shows the comparison between advance payments received and remaining payments. It helps in understanding how much payment is completed and how much is still pending.

The overall layout is clean and organized, making it easy for users to quickly analyze important information and make better decisions.

```
const getFilteredPayments = () => {
  const now = new Date();
  let payments = [];
  orders.forEach(order => {
    if (order.paymentHistory) {
      order.paymentHistory.forEach(p => {
        payments.push({ ...p, order });
      });
    }
  });
  return payments.filter(p => {
    const d = new Date(p.date);
    if (view === "daily") return d.toDateString() === now.toDateString();
    if (view === "weekly") {
```

```
      const weekAgo = new Date();
      weekAgo.setDate(now.getDate() - 7);
      return d >= weekAgo;
    }
    if (view === "monthly") {
      const monthAgo = new Date();
      monthAgo.setMonth(now.getMonth() - 1);
      return d >= monthAgo;
    }
    return true;
  });
};
```

6.15 GENERATED REPORT



Gurumauli Aqua

Water Jar Delivery System

Delivery & Payment Report

Type: MONTHLY

Generated: 4/6/2026, 7:59:30 AM

Fully Paid Deliveries

Order ID	Customer	Jars	Amount	Status
ORD-e620	Hiteshree Desale	50	1000	Paid
ORD-1fce	Pravin Desale	30	600	Paid

Pending Deliveries

Order ID	Customer	Jars	Status	Pending Amount
ORD-6e28	Khushbu Patil	20	Confirmed	320
ORD-6e1c	Khushbu Patil	50	Pending	0
ORD-d0cf	Manasi Patil	80	Awaiting Payment	0
ORD-5075	Hiteshree Desale	2	Pending	0
ORD-e617	Hiteshree Desale	1	Awaiting Payment	0
ORD-e611	Hiteshree Desale	50	Pending Delivery	800
ORD-202b	Parinita Desale	25	Pending Delivery	400
ORD-9afe	ketal patil	50	Pending Delivery	800
ORD-cf3f	Pravin Desale	153	Pending Delivery	2448

Figure 6.15: Report

This report shows the summary of delivery and payment details for a selected period(monthly).

It provides information about fully paid deliveries and pending deliveries with their payment status.

The report includes:

- List of fully paid orders with amount and status
- List of pending orders with remaining payment details
- Order details like customer name, number of jars, and status

It helps the supplier to track completed payments and identify pending payments easily.

6.16 DELIVERY STAFF MANAGEMENT (SUPPLIER)

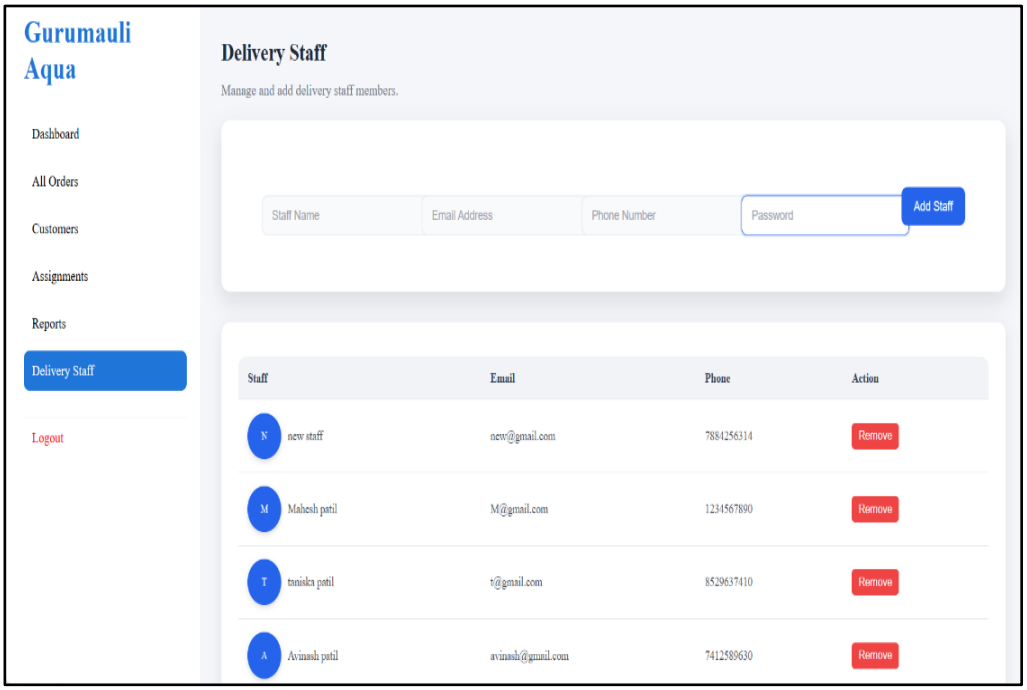


Figure 6.15: Delivery Staff Management Page

This page is used to manage delivery staff in the Water Jar Distribution System. It allows the supplier to handle all delivery staff activities in one place, such as adding, viewing, and removing staff members. The supplier can add new staff by entering details like name, email, phone number, and password, which are stored in the system.

The page also displays a list of all delivery staff with their details for easy tracking. If any staff member is no longer needed, they can be removed easily. This helps in keeping the delivery team organized and ensures smooth delivery operations.

Overall, this page improves the management of delivery staff and ensures smooth delivery operations by maintaining proper records and easy access to staff information.

Supplier can:

- Add new delivery staff by entering name, email, phone and password
- View list of all delivery staff members
- Check staff details (name, email, phone number)

- Remove delivery staff if not required
- Manage delivery team efficiently

6.17 CUSTOMER LIST PAGE

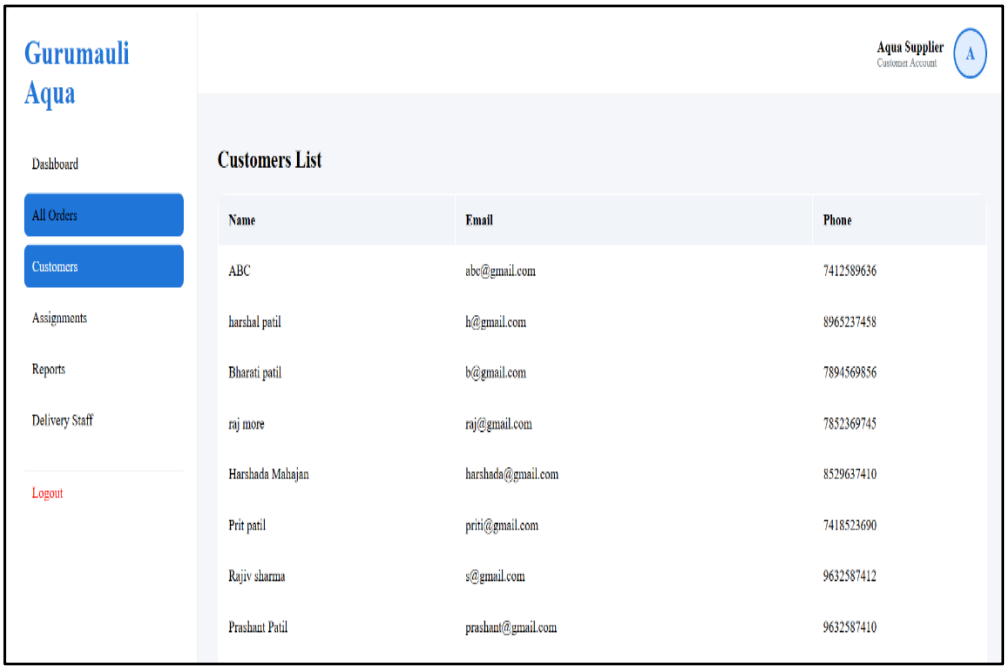


Figure 6.17: Customer List Page

This page displays the list of all customers in the Water Jar Distribution System. It helps the supplier to view and manage customer details in an organized way. The page shows important information such as customer name, email, and phone number, making it easy to identify and contact customers.

The supplier can view all registered customers in a structured list format. This makes it easier to track customer information and maintain proper records. The page is designed to be simple and user-friendly so that the supplier can quickly access and manage customer data without any confusion.

Supplier can:

- View all registered customers
- Check customer details (name, email, phone)
- Manage customer information easily

6.18 DELIVERY STAFF DASHBOARD

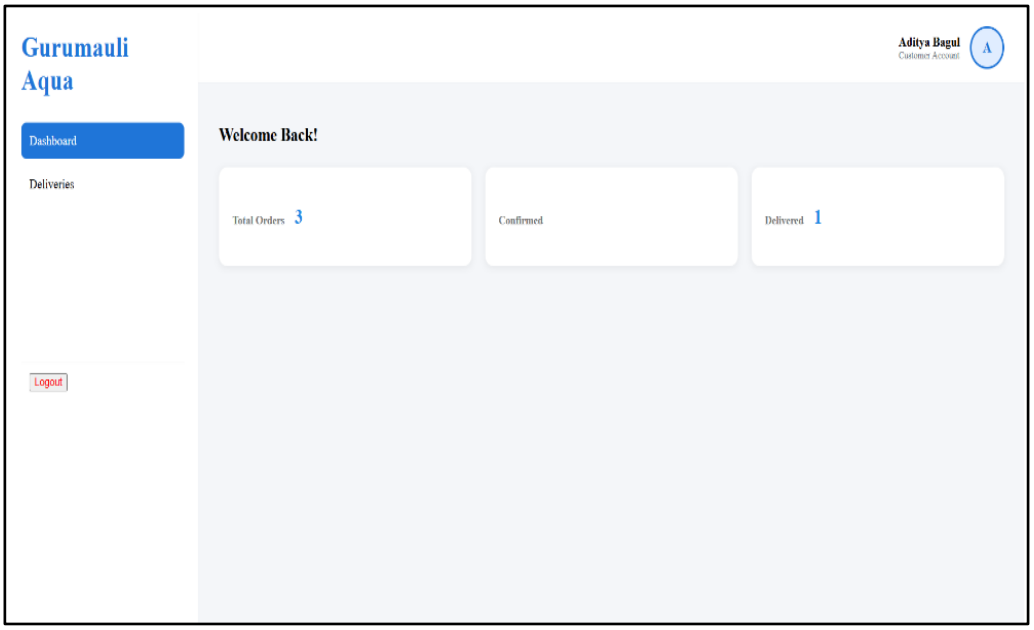


Figure 6.18: Delivery Staff Dashboard

The Delivery Staff Dashboard includes a header and a menu on the left side. The menu provides options such as Dashboard, Deliveries, and Logout for easy navigation. The main dashboard displays summary cards that show important information like total assigned orders, confirmed orders, and delivered orders.

This helps delivery staff quickly understand their work status. From this dashboard, delivery staff can view their assigned delivery tasks, track the current delivery status, and monitor completed deliveries. The page is designed to be simple and user-friendly, making it easy to manage delivery operations efficiently.

The dashboard shows summary cards which include:

- Total assigned orders
- Confirmed orders
- Delivered orders

Delivery staff can:

- View assigned delivery tasks
- Track delivery status
- Monitor completed deliveries

6.19 ASSIGNED DELIVERIES PAGE

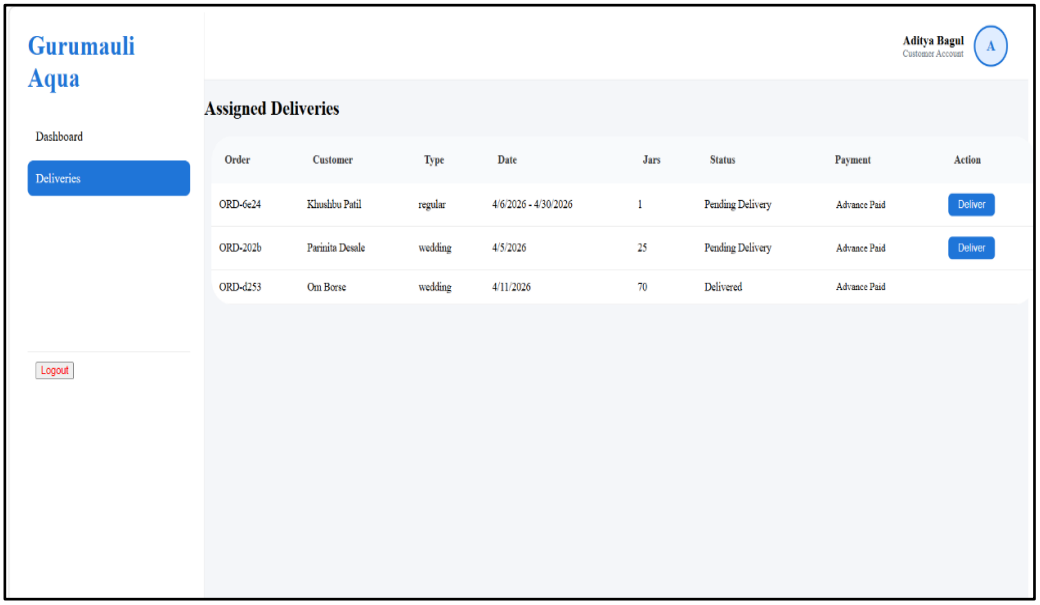


Figure 6.19: Assigned Deliveries Details page

This page shows all deliveries assigned to the delivery staff. It helps them track delivery details and update delivery status. Assigned Deliveries page contains header and menu at left side. This page displays assigned deliveries in table format.

The table shows:

- Order ID
- Customer name
- Order type
- Delivery date
- Number of jars
- Delivery status
- Payment status

Delivery staff can:

- View all assigned deliveries
- Check delivery and payment details
- Mark order as delivered using “Deliver” button
- Track pending and completed deliveries

Status labels are used as follows:

1. Pending Delivery – Delivery not completed

2. Delivered – Delivery completed

```
const markDelivered = async (id) => {
  await API.put(`/delivery/deliver/${id}`);

  fetchOrders();
};

const collectEmpty = async (id) => {

  await API.put(`/delivery/collect/${id}`, {
    collected: 1
  });

  fetchOrders();
};
```

CHAPTER 7
TESTING

This chapter presents the testing and validation process carried out for the Web Application for Water Jar Distribution System. The primary aim of testing is to ensure that the system functions correctly, efficiently, and meets all specified requirements. Testing was performed on three major modules of the system: Customer Login, User Registration, and New Delivery Request (Order Form – Core Functionality). The login module acts as the entry point for users and ensures secure authentication. The registration module allows new users to create accounts with proper validation. The order module handles the main functionality of placing water jar delivery requests by selecting order type, number of jars, dates, and delivery details.

The objective of this testing phase is to verify that all modules work accurately for valid inputs and appropriately handle invalid or incomplete inputs. The system was tested for various scenarios including successful login, invalid credentials, empty fields, valid registration, duplicate user data, valid order placement, incorrect order details, and validation checks for input fields.

Both positive and negative testing approaches were used to ensure the reliability and robustness of the system. Positive testing verifies correct behavior with valid inputs, whereas negative testing checks how the system responds to incorrect or unexpected inputs.

Testing was carried out using manual methods as well as automation tools such as Selenium IDE and Selenium WebDriver improve efficiency and accuracy.

7.1 TEST PLAN

The test plan defines the objectives, scope, testing environment, tools, entry and exit criteria, and risk factors associated with the testing cycle of the Web Application for Water Jar Distribution System.

This test plan has been prepared following the IEEE standard format to ensure a well-structured and standardized documentation approach. It focuses on testing the core modules of the system such as User Registration, Customer Login, and Order Form.

A structured test plan ensures that testing activities are disciplined, consistent, and reproducible, and are aligned with the overall quality goals of the project.

Test Plan: (Web Application for Water Jar Distribution System)

Version: 1.0

Test Plan ID: TP_WJDS_01

1. Introduction

This test plan defines the testing strategy for the Login, User Registration, and Order Form modules of the Web Application for Water Jar Distribution System. The purpose of testing is to ensure that all modules function correctly with valid inputs and handle invalid inputs properly. It also ensures system reliability, usability, and security.

2. Test Items

- Login module (email, password, authentication)
- Registration module (user details validation and creation)
- Order form module (delivery request, address, jar quantity)
- UI elements (forms, buttons, error messages)
- Backend validation and database operations

3. Features to be Tested

- Valid and invalid login scenarios
- Empty field validation (login & registration)
- Email format validation
- Password validation
- User registration with valid/invalid data
- Duplicate registration handling
- Order form submission
- Date validation (start & end date)
- Address and pincode validation
- Total price calculation
- Role-based redirection

4. Features Not to be Tested

- Payment Module
- Inventory management

5. Approach

- Functional and non-functional testing will be performed
- Both positive and negative test cases will be executed
- Manual testing will be used primarily
- Selenium automation can be used for repetitive tasks

- UI and usability testing will be included

6. **Pass / Fail Criteria**
    - All critical test cases must pass
    - Expected results should match actual results
    - No critical or blocking defects should remain
    - All major defects must be fixed and retested
    - Minimum 95% test coverage should be achieved
7. **Suspension Criteria and Resumption Requirements**
    - Testing will be stopped if critical bugs are found.
    - Testing will resume after bug fixing and system stabilization.
8. **Test Deliverables**
    - Test plan document
    - Test cases document
    - Defect report
    - Test summary report
    - Automation scripts
9. **Test Tasks**
    - Prepare test plan and test cases
    - Execute test cases
    - Identify and log defects
    - Retest fixed defects
    - Prepare final report
10. **Test Environment**
    - Operating System: Windows 10/11
    - Browser: Google Chrome, Microsoft Edge
    - Frontend: React (localhost)
    - Backend: Node.js + Express.js
    - Database: MongoDB
    - Server: Localhost
    - Tools: Selenium IDE, Selenium WebDriver
11. **Responsibilities**
    - Test Engineer: Execute tests cases and report defects
    - Developer: Fix defects
    - Project Guide: Review testing process

12. Staffing and Training

Testing activities are performed during the final phase of project development.

13. Scheduling

Project deadline: 20 March 2026, 12:00 PM

14. Risks

- Incomplete requirements
- Browser compatibility issues

15. Approvals

Project Guide and Team Members

7.2 TEST CASES

- **Registration Functionality**

Table 7.1: Test Case (Registration Functionality)

Test Case ID	Objective	Input (Steps)	Expected Result	Actual Result	Status
TC_REG_001	Registration with valid data	Name: abc Email: abc@gamil.com Mobile: 1234567897 Password: abc@123 Role: Customer	User should register successfully and data saved in database	User registered successfully and data saved	Pass
TC_REG_002	Registration with empty field	All fields blank	System should display "Please fill all fields"	Message displayed.	Pass
TC_REG_003	Registration with invalid email.	Email:abc123	Show should display "invalid email format"	Error message displayed.	Pass
TC_REG_004	Registration with empty password.	Password:	System Should Display "password is required"	Message displayed	Pass

Test Case ID	Objective	Input (Steps)	Expected Result	Actual Result	Status
TC_REG_005	Registration with weak password	Password: 12345	System Should display "Weak password"	Message displayed correctly	Pass
TC_REG_ 006	Registration with duplicate email	Email already registered	System should display "Email already exists"	Error message displayed	Pass
TC_REG_007	Registration with invalid mobile number	Mobile:1234	System should "Invalid mobile number"	Error Message displayed	Pass
TC_REG_008	Role selection validation	No role selected	System should display "Please select role"	Message displayed	Pass
TC_REG_009	Registration as Customer	Valid details+ Role: Customer	User should register Successfully	Registered successfully	Pass
TC_REG_010	Multiple registration attempts restriction	Enter invalid/duplicate data multiple times and click Register repeatedly.	System should restrict multiple registration attempts.	System allows unlimited registration attempts without any restriction	Fail

- **Login Functionality**

Table 7.2: Test Case for Login Functionality

Test Case ID	Objective	Input (Steps)	Expected Result	Actual Result	Status
TC_LGN_001	Login with valid credentials	Email:abc@gamil.com Password: abc@123 Click Login	User should login successfully And redirect to dashboard	User login in and redirected	Pass
TC_LGN_002	Login with invalid password	Enter valid email Password:wrong	System should display "lnvalid credentials"	Error message displayed.	Pass
TC_LGN_003	Login with invalid email.	Email:wrong Password: correct	Show should display "User not found"	Error messag e display ed .	Pass
TC_LGN_004	Login with empty field	Email: blank Password: balnk	System Should Display "Please fill all fields"	Message displayed	Pass
_LGN_005	Login with empty password	Enter email Password: blank	System Should display "Password is required"	Message displayed correctly	Pass

Test Case ID	Objective	Input (Steps)	Expected Result	Actual Result	Status
TC_LGN_ 006	Login with empty email	Email : blank Password: entered	System should display "Email is required"	Error message displayed	Pass
TC_LGN_007	Invalid email format	Email: admin123 Password: correct	System should display "Invalid email format"	Error Message displayed	Pass
TC_LGN_008	Login button functionality	Click Login button with valid inputs	System should Process login	Login successful	Pass
TC_LGN_009	Verify forgot Password functionality	Click forgot password +enter email	System should send reset link	Reset link not sent	Fail
TC_LGN_010	Multiple failed login attempts restriction	Enter wrong password multiple times	System should restrict login or show warning	System allows unlimited attempt	Fail

- **Test Cases:- Order Form**

Table 7.3: Test Case for Order Form

Test Case ID	Objective	Input (Steps)	Expected Result	Actual Result	Status
TC_ORD_001	Order with valid details	Select Order Type: Regular Jars 2 Valid start & End Date Address filled Pincode:422001	Order should be placed successfully	Order placed successfully	Pass
TC_ORD_002	Empty fields Validation	Leave all fields blank and click Place Order	System should display "Please fill all fields"	Message displayed.	Pass
TC_ORD_003	Invalid pincode	Enter piccode:123	Show should show"invalid pincode"	Error message displayed.	Pass
TC_ORD_004	Invalid date selection	End date before start date	System Should show"error message"	Error displayed	Pass
TC_ORD_005	Address empty	Leave address blank	System should show "Address required"	Message displayed	Pass

Test Case ID	Objective	Input (Steps)	Expected Result	Actual Result	Status
TC_ORD_ 006	Jar quantity validation	Enter jars:0	System should not allow order	Error displayed	Pass
TC_ORD_007	Price calculation	Jars:2 (Rs 20 each)	Total should be Rs 40	Correct total show	Pass
TC_ORD_008	Advance calculation	Total Rs 40	Advance Should be Rs 8(20%)	Correct advance shown	Pass
TC_ORD_009	Order type selection	Select Event Order	System should accept selection	Order type selected	Pass
TC_ORD_010	Multiple order attempts	Click Place Order multiple times	System should redirect multiple requests	System allows multiple Orders without restriction	Fail

7.3 DEFECT REPORT

The defect report is a crucial document used to record and track defects identified during the testing process of the Web Application for Water Jar Distribution System. It provides detailed information about each defect, including its description, severity, priority, steps to reproduce, and current status.

This report helps the development and testing teams to understand, prioritize, and resolve issues efficiently. It also ensures effective communication among team members and maintains transparency throughout the defect management process.

- **Defect Report (Registration –Multiple Attempts Restriction)**

Table 7.4: Defect Report for Registration

Defect ID:	DR_REG_001
Project Name:	Web Application for Water Jar Distribution System
Product:	Web Application
Release Version:	v1.0
Module:	User Registration Module
Detected Build Version:	v1.1
Summary:	System allow unlimited registration attempts without restriction
Description:	While testing the registration module, it was observed that the system does not restrict multiple registration attempts. When the user enters invalid or duplicate data multiple times and repeatedly clicks on the register button, the system allows unlimited attempts without displaying any warning message or blocking mechanism. This may lead to misuse of the system and potential security risks.
Steps to Reproduce	Open registration page Enter invalid or duplicate data Click on Register button multiple times

Expected Result:	System should restrict multiple registration attempts or display warning message for security
Actual Result:	System allows unlimited registration attempts without any restriction
Severity:	High (Security risk due to unlimited attempts)
Priority:	High (Need immediate fix)
Reported By:	Test Engineer
Status:	Open

- **Defect Report :- (Login – Forgot Password)**

Table 7.5: Defect Report for Login –Forgot Password

Defect ID:	DR_LGN_002
Project Name:	Web Application for Water Jar Distribution System
Product:	Web Application
Release Version:	v1.0
Module:	Customer Login Module-Forgot Password
Detected Build Version:	v1.1
Test Case ID	TC_LGN_010
Summary:	Password reset link is not sent to registered email.

Description:	While testing the Forgot Password functionality, it was observed that when a valid registered email is entered, the system does not send any password reset link. Additionally, no success or error message is displayed. This prevents the user from resetting the password and accessing the account, indicating a failure in backend or email service integration.
Steps Reproduce:	Open login page Click on "Forgot Password" Enter valid registered email Click on submit/reset button
Expected Result:	System should send a password reset link to the registered email
Actual Result:	No reset link is sent a password reset link to the registered email
Severity:	High (User cannot reset password and access account)
Priority:	High (Needs immediate fix)
Reported By:	Test Engineer
Status:	Open

- **Defect Report :- (Order Form –Multiple Order Attempts)**

Table 7.6: Defect Report for Order Form

Defect ID:	DR_ORD_003
Project Name:	Web Application for Water Jar Distribution System
Product:	Web Application
Release Version:	v1.0
Module:	Order Module
Detected Build Version:	v1.1
Test Case ID	TC_ORD_010
Summary:	System allows multiple order submission without restriction
Description:	While testing the order module, it was observed that when the user clicks on the "Place Order" button multiple times, the system processes multiple order requests without any restriction or validation. There is no mechanism to prevent duplicate order submissions or display a warning message. This may result in duplicate orders, incorrect billing, and system misuse.
Steps to Reproduce:	Open order page Select product/details Click on "Place Order" button multiple times
Expected Result:	System should restrict multiple order requests or allow only one order submission with proper validation
Actual Result:	System allows multiple orders without any restriction

Severity:	High (May lead to duplicate orders and billing issues)
Priority:	High (Needs immediate fix)
Reported By:	Test Engineer
Status:	Open

7.4 AUTOMATION TESTING (SELENIUM)

Automation testing is performed to validate the functionality of the Web Application for Water Jar Distribution System using automated tools. It helps in reducing manual effort, improving accuracy, and ensuring consistency in repetitive test execution.

In this project, automation testing is carried out using Selenium IDE and Selenium WebDriver. These tools are used to automate the testing of the Login module.

Selenium IDE is a record-and-playback tool that allows testers to create automated test cases without programming knowledge. It is mainly used to record user actions such as entering login credentials, clicking buttons, and navigating between pages. The recorded test scripts can be executed multiple times to verify system behavior.

Selenium WebDriver is an advanced automation testing tool that allows interaction with web browsers through programming. It is used to write scripts that simulate user actions such as typing input, clicking elements, and validating outputs. WebDriver supports multiple browsers and provides better control for handling complex test scenarios.

Automation was performed by recording login scenarios such as valid login, invalid login, and empty field validation using Selenium IDE.

The automated test scripts were executed multiple times to ensure consistent results. The results obtained from automation testing matched the expected outcomes for most test cases, except for certain failed scenarios such as forgot password functionality not working properly.

7.4.1 Selenium Ide Execution

In this project, Selenium IDE was used to automate the Customer Login, Supplier Login, and Delivery Staff Login functionalities. The recorded scripts were executed multiple times to verify consistent system behavior. The execution results showed that most of the test cases passed successfully, while some scenarios such as forgot password functionality failed.

- **Login Functionalities Selenium :-**

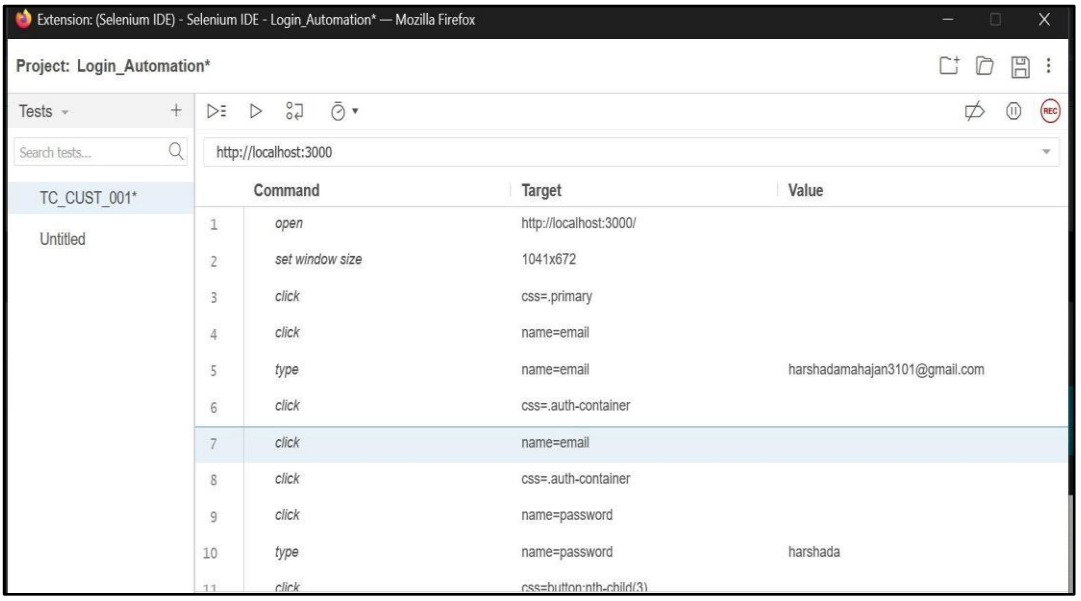


Figure 7.1 : Customer Login (Valid Email & Valid Password)

This screenshot shows the execution of the customer login test case using valid email and valid password in Selenium IDE. The automation script enters correct customer credentials and performs the login action. The system successfully authenticates the user and redirects to the customer dashboard. This confirms that the login functionality works correctly for valid inputs and allows authorized access to the system.

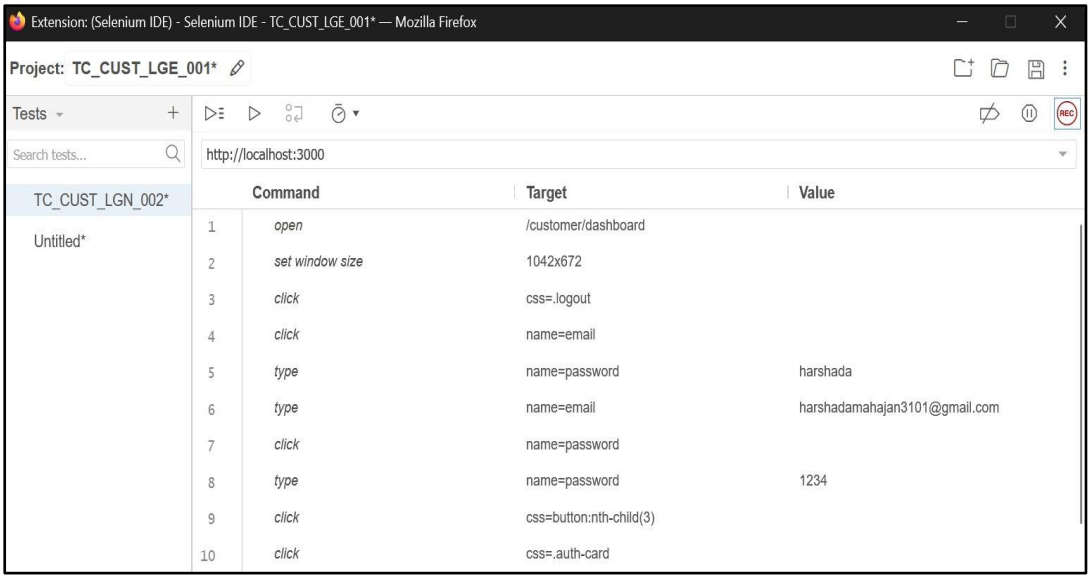


Figure 7.2 : Invalid Password (Valid Email & Invalid Password)

This screenshot illustrates the execution of the customer login test case with a valid email and an invalid password. The automation script attempts to log in using incorrect password credentials. The system detects the mismatch and displays an appropriate error message such as "Invalid credentials" while preventing access. This verifies proper password validation and ensures that unauthorized users cannot log in to the system.

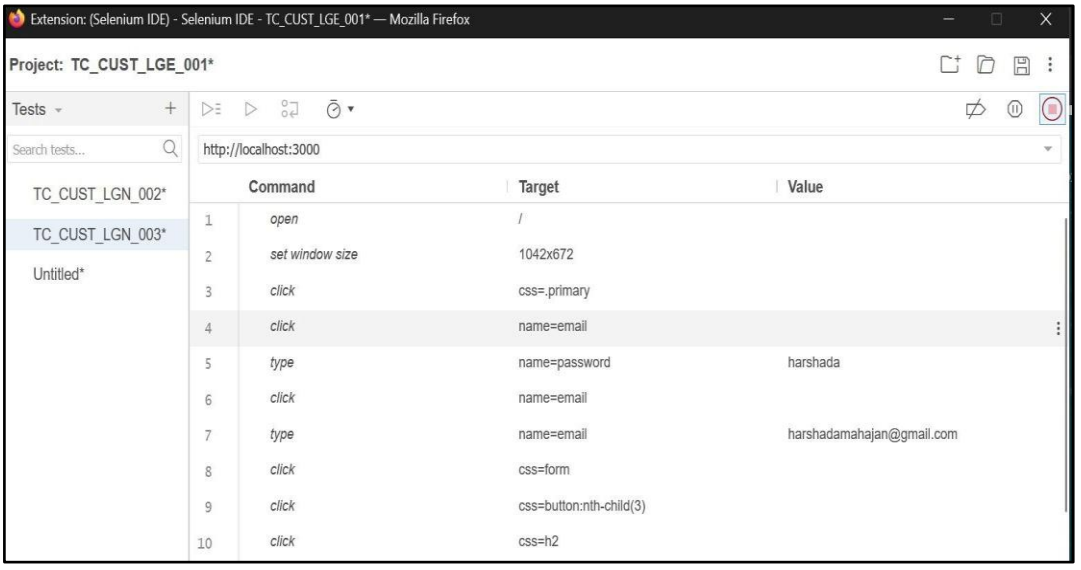


Figure 7.3 : Invalid Email (Invalid Email & Valid Password)

This screenshot illustrates the execution of the customer login test case with an invalid email and a valid password. The automation script attempts to log in using an email that is not registered in the system. The system identifies that the email is incorrect and displays an appropriate error message such as "Invalid credentials". The login attempt is denied, ensuring that only registered users can access the system.

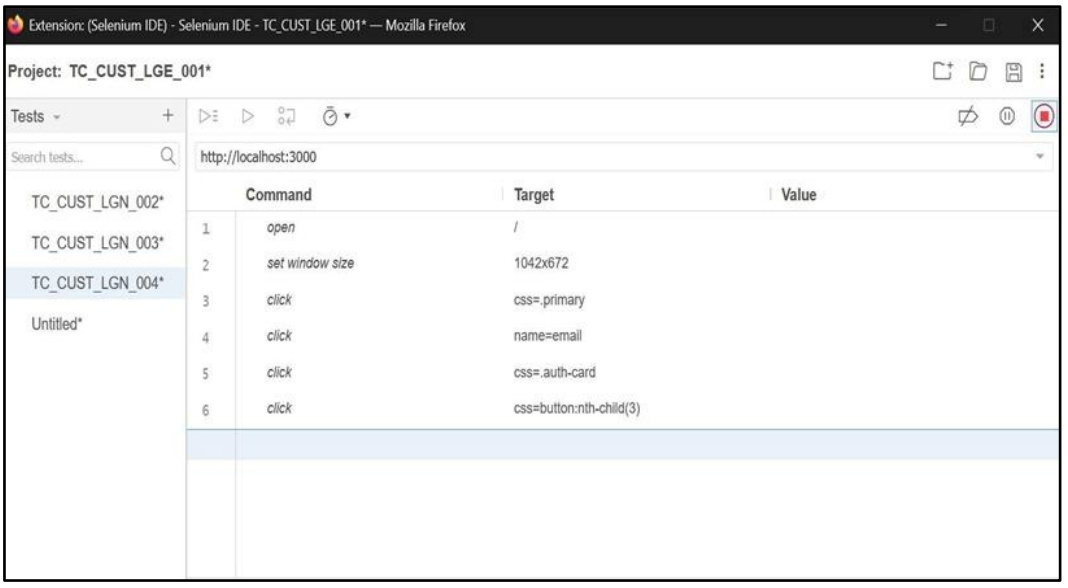


Figure7.4: Login with empty fields (No Email & No Password)

This screenshot illustrates the execution of the login test case with empty input fields, where both email and password are not entered. The automation script attempts to perform the login action without providing any credentials. The system validates the input fields and displays an appropriate message such as "Please fill in all fields", preventing the login process. This confirms that the system correctly handles empty field validation and ensures that users cannot proceed without entering required information.

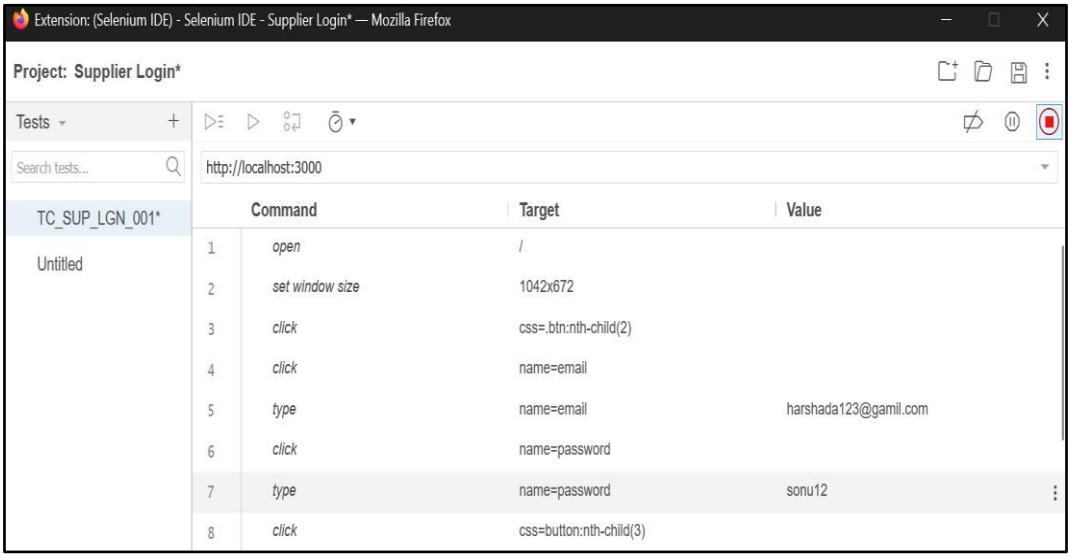


Figure 7.5 : Supplier Login (Valid Email & Valid Password)

This screenshot illustrates the execution of the supplier login test case using valid email and valid password credentials. The automation script enters correct supplier login details and performs the login action. The system successfully

authenticates the supplier and redirects to the supplier dashboard. This confirms that the system behaves correctly for valid inputs and allows authorized supplier access.

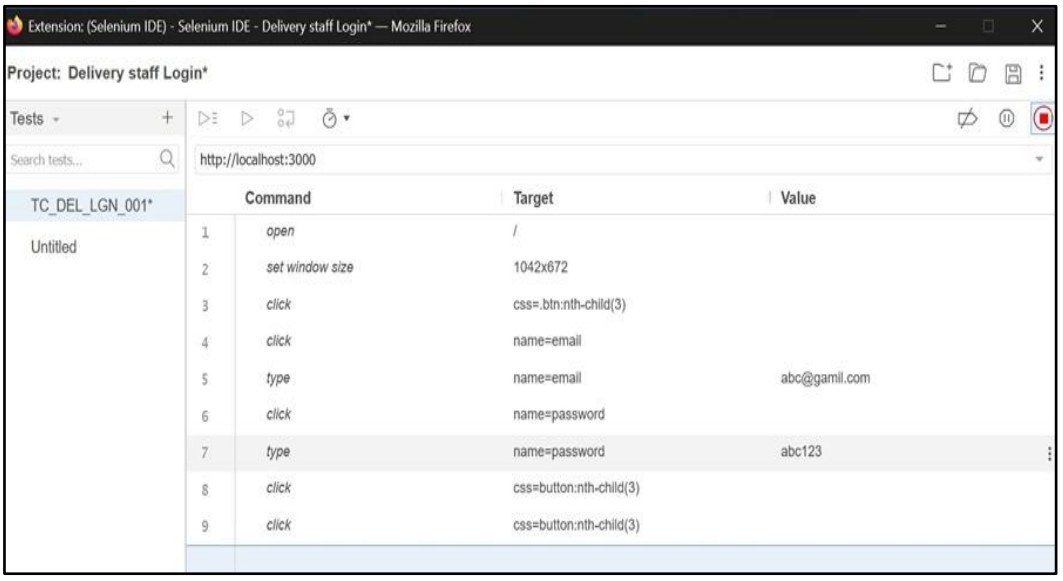


Figure 7.6 : Delivery Staff Login (Valid Email & Valid Password)

This screenshot illustrates the execution of the delivery staff login test case using valid email and valid password credentials. The automation script enters correct delivery staff login details and performs the login action. The system successfully authenticates the delivery staff and redirects to the delivery dashboard. This confirms that the system behaves correctly for valid inputs and allows authorized delivery staff access.

7.4.2 Selenium Webdriver Execution

Selenium WebDriver is used to automate testing by interacting directly with the web browser using scripts. It simulates user actions such as entering inputs, clicking buttons, and navigating between pages. In this project, WebDriver was used to automate the Login functionality for Customer, Supplier, and Delivery Staff. It provides better control and flexibility compared to Selenium IDE.

For the Login module, WebDriver automates entering email and password, clicking the login button, and verifying successful login or error messages. It also checks role-based redirection.

WebDriver also validates different test scenarios such as valid login, invalid credentials, and empty field validation for all user roles. It ensures that appropriate messages are displayed and unauthorized access is restricted. This helps in

verifying that the system behaves correctly for both valid and invalid inputs.

- **WebDriver Script (Login)**

```
package test;
import java.time.Duration;
import org.openqa.selenium By;
import org.openqa.selenium.WebDriver;
importorg.openqa.selenium.firefox.FirefoxDriver; public class LoginTestCases {
// ═══════════════ DRIVER SETUP - Common method ═══════════════
public static WebDriver setupDriver() throws InterruptedException{
System.setProperty("webdriver.gecko.driver","C:\\Selenium\\geckodriver.exe");
WebDriver driver = new FirefoxDriver();
driver.manage().window().maximize();
driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(5));
driver.get("http://localhost:3000/");
Thread.sleep(2000); return driver;
}
// ═══════════════ TC_LGN_001 - Valid Customer Login ═══════════════
public static void tc001_CustomerLogin() throws InterruptedException {
WebDriver driver = setupDriver(); System.out.println("TC_LGN_001 - Valid
Customer Login"); login(driver, "harshada@gmail.com", "123456");
Thread.sleep(3000);
if (driver.getCurrentUrl().contains("customer-dashboard"))
System.out.println("TC_LGN_001 - PASS");
else
System.out.println("TC_LGN_001 - FAIL");          driver.quit();  }
// ═══════════════ TC_LGN_002 - Valid Supplier Login ═══════════════
public static void tc002_SupplierLogin() throws InterruptedException {
WebDriver driver = setupDriver(); System.out.println("TC_LGN_002 - Valid
Supplier Login");
login(driver, "supplier@gmail.com", "123456"); Thread.sleep(3000);
if (driver.getCurrentUrl().contains("supplier-dashboard"))
System.out.println("TC_LGN_002 - PASS");
else
```

```
System.out.println("TC_LGN_002 - FAIL"); driver.quit();
}
// ═══════════ TC_LGN_003 - Valid Delivery Staff Login ═══════════
public static void tc003_DeliveryLogin() throws InterruptedException {
 WebDriver driver = setupDriver(); System.out.println("TC_LGN_003 - Valid
    Delivery Login"); login(driver, "delivery@gmail.com", "123456");
    Thread.sleep(3000);
if(driver.getCurrentUrl().contains("delivery-dashboard"))
   System.out.println("TC_LGN_003 - PASS");
  else
System.out.println("TC_LGN_003 - FAIL"); driver.quit();}
// ═══════════TC_LGN_004 - Invalid Password ═══════════
   public static void tc004_InvalidPassword() throws InterruptedException {
  WebDriver driver = setupDriver();
 System.out.println("TC_LGN_004- Invalid Password");
 login(driver, "harshada@gmail.com", "wrong123"); Thread.sleep(2000);
System.out.println("TC_LGN_004 PASS"); driver.quit();
}
// ═══════════ TC_LGN_005 - Invalid Email ═══════════
    public static void tc005_InvalidEmail() throws InterruptedException {
WebDriver driver = setupDriver();
     System.out.println("TC_LGN_005 - Invalid Email");
login(driver, "wrong@gmail.com", "123456"); Thread.sleep(2000);
System.out.println("TC_LGN_005 - PASS");
 driver.quit();
}
// ═══════════ TC_LGN_006 - Empty Fields ═══════════
public static void tc006_EmptyFields() throws InterruptedException { WebDriver
driver = setupDriver();
System.out.println("TC_LGN_006 - Empty Fields");   clearFields(driver);
driver.findElement(By.cssSelector("button[type='submit']")).click();
Thread.sleep(2000);
 System.out.println("TC_LGN_006 - PASS");  driver.quit();
```

}

- **Execution for Login:-**

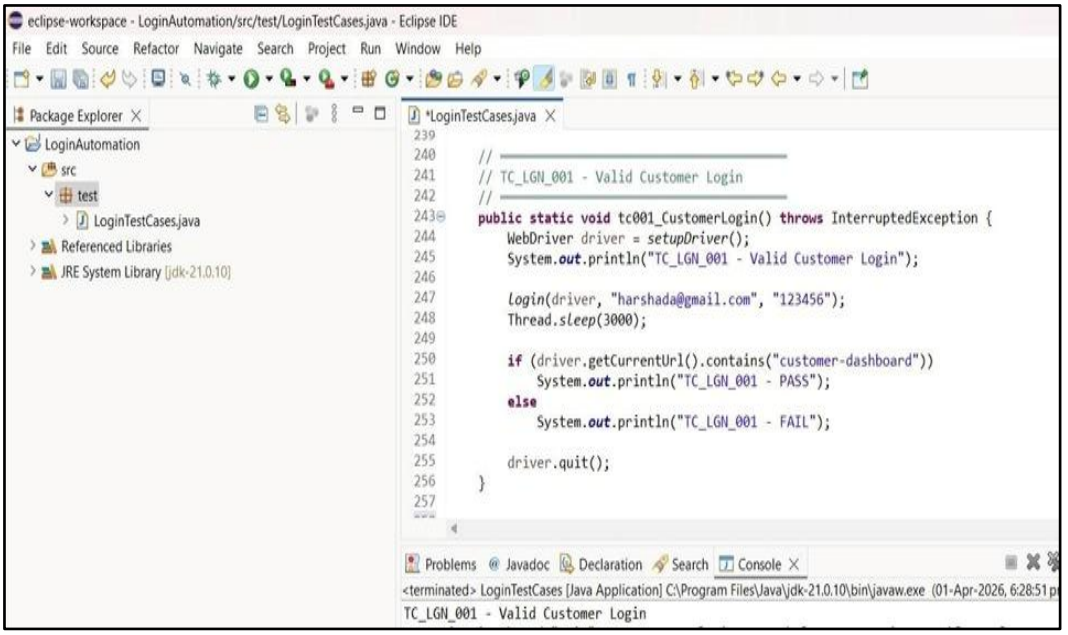


Figure 7.7: Valid Customer Login

This screenshot shows the execution of the Selenium WebDriver script for valid customer login. The script enters correct customer email and password and performs the login action. The system successfully authenticates the user and redirects to the Customer Dashboard. This confirms correct login functionality and role-based navigation.

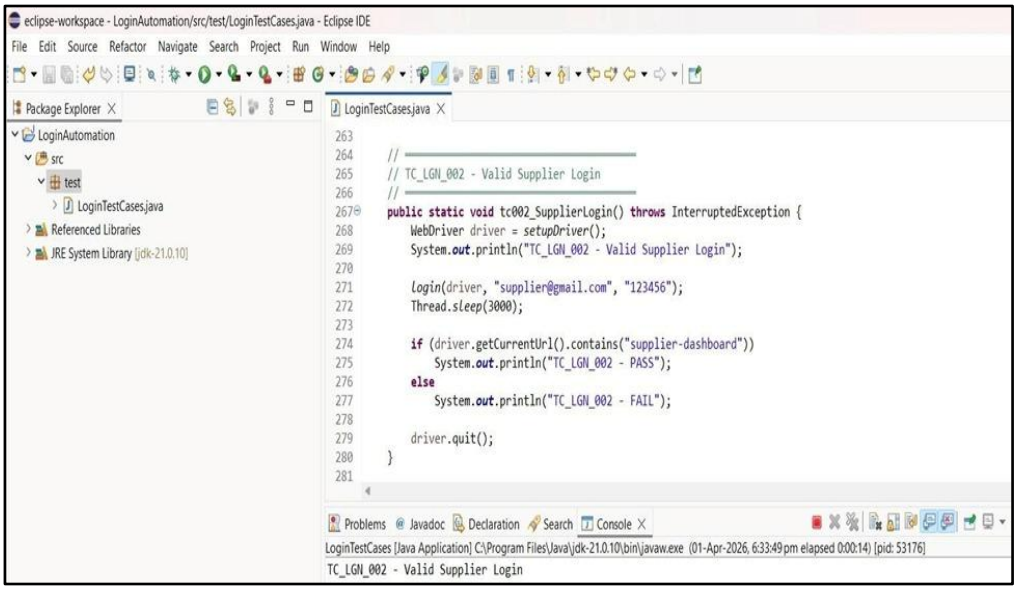


Figure 7.8: Valid Supplier Login

This screenshot shows the execution of the Selenium WebDriver script for valid supplier login. The script enters correct supplier email and password and performs the login action. The system successfully authenticates the supplier and redirects to the Supplier Dashboard. This confirms correct login functionality and role-based navigation.

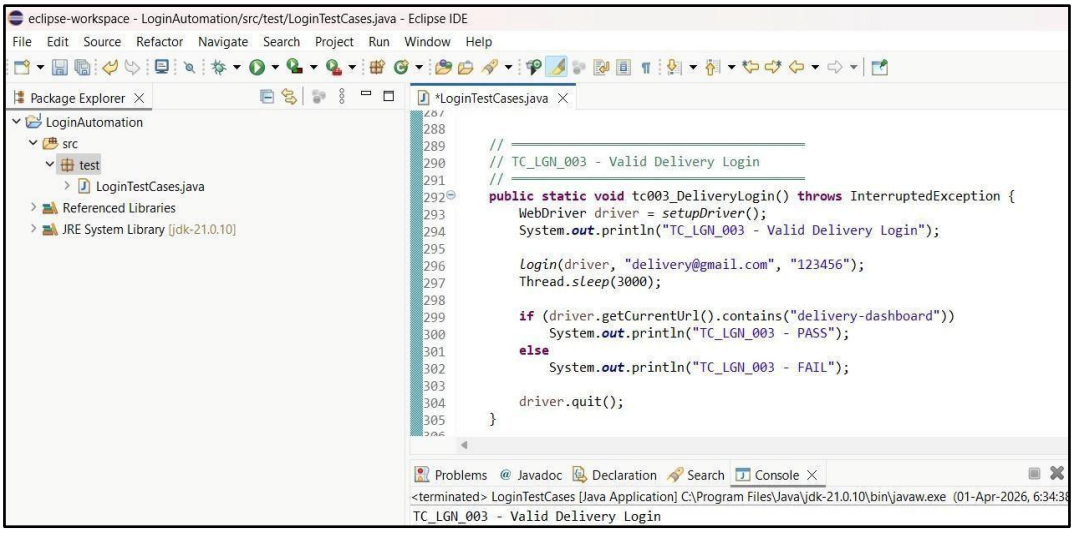


Figure 7.9: Valid Delivery Login

This screenshot shows the execution of the Selenium WebDriver script for valid delivery staff login. The script enters correct delivery staff email and password and performs the login action. The system successfully authenticates the delivery staff and redirects to the Delivery Dashboard. This confirms correct login functionality and role- based navigation.

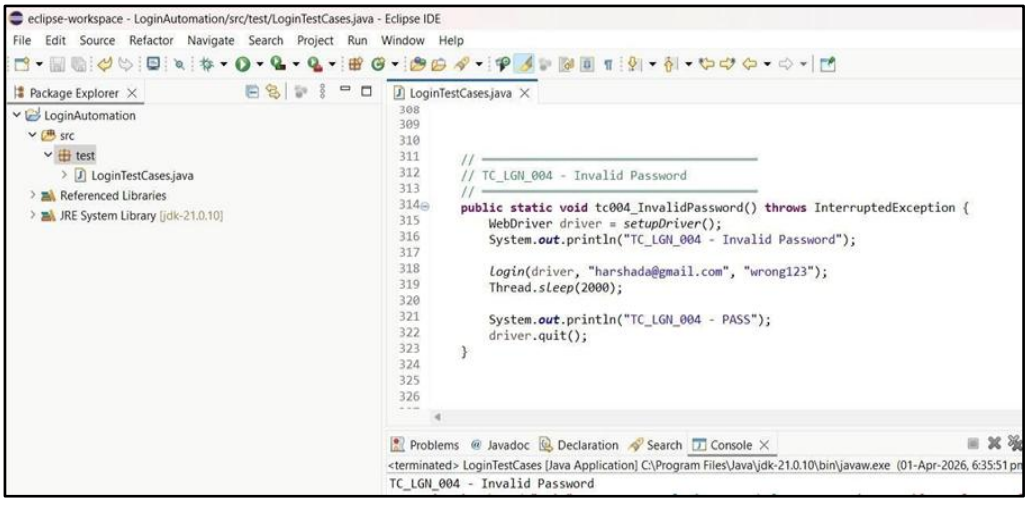


Figure 7.10: Invalid Password

This screenshot shows the execution of the Selenium WebDriver script for login with an invalid password. The script enters a valid registered email and an incorrect password and performs the login action. The system detects the mismatch in credentials and displays an appropriate error message such as "Invalid password", preventing the user from logging in. This confirms proper validation of login credentials and ensures system security

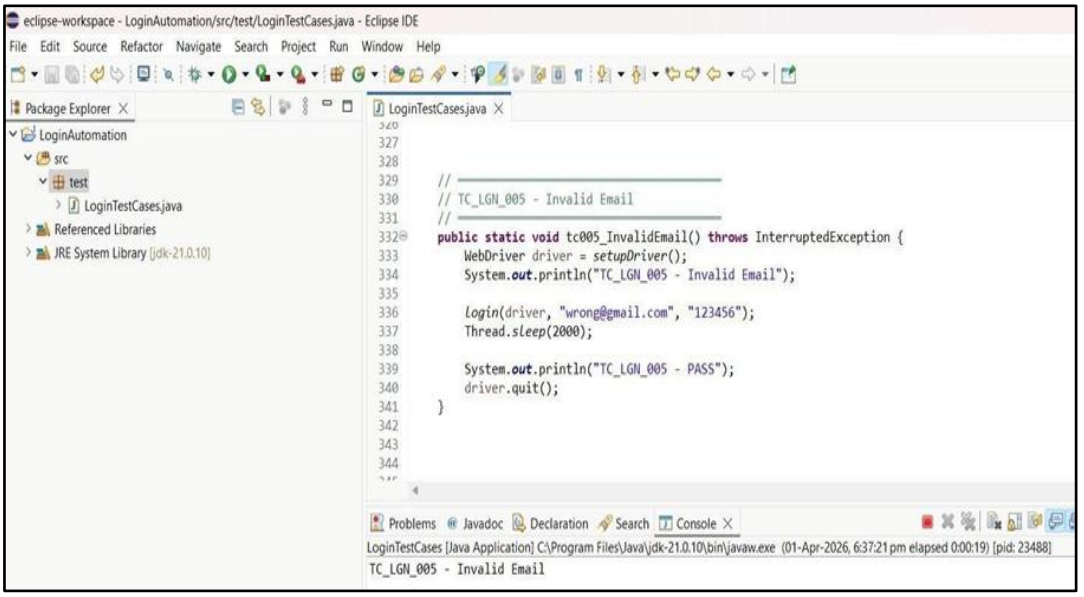


Figure 7.11: Invalid Email

This screenshot shows the execution of the Selenium WebDriver script for login with an invalid email. The script enters an incorrect or unregistered email along with a valid password and performs the login action. The system identifies that the email is not registered or invalid and displays an appropriate error message such as "Invalid email", preventing the user from logging in. This confirms proper validation of login credentials and ensures system security.

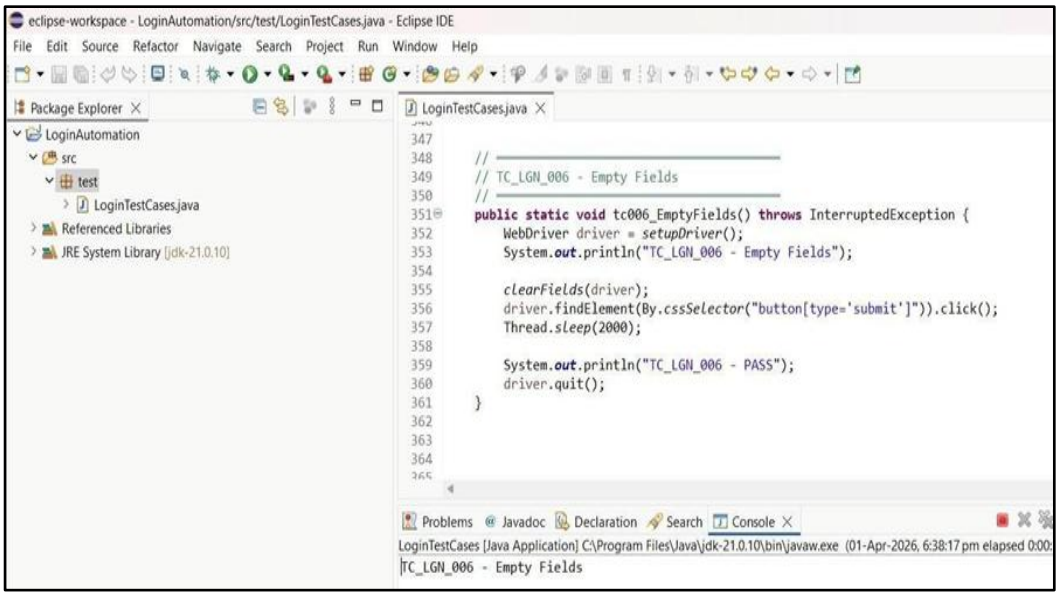


Figure 7.12: Empty Fields

This screenshot shows the execution of the Selenium WebDriver script for login with empty input fields. In this test case, the script attempts to perform the login operation without entering any email and password. The system detects that the required input fields are left blank and displays a validation message such as "Please fill in all fields", thereby preventing the login process. This behavior ensures that the system enforces mandatory field validation and does not allow users to proceed with incomplete input. It confirms the reliability of the login functionality and enhances overall system security.

7.5 TEST SUMMARY

The test summary provides an overview of the testing activities performed on the Web Application for Water Jar Distribution System. The testing process was carried out mainly on the Login functionality, which includes Customer Login, Supplier Login, and Delivery Staff Login.

A total of multiple test cases were designed and executed to validate the login module. The test cases covered various scenarios including valid login, invalid password, invalid email, empty field validation, and user interface validation. Both positive and negative testing approaches were used to ensure system reliability and robustness.

The majority of test cases were executed successfully, and the system performed as expected under normal conditions. The login module successfully authenticated valid users such as customers, suppliers, and delivery staff, and redirected them to their respective dashboards. It also handled invalid inputs efficiently by displaying appropriate error messages.

However, some test cases failed during execution. Issues were identified in certain scenarios such as improper validation messages and handling of edge cases in the login module. These defects were documented and reported for further resolution.

Automation testing using Selenium IDE and Selenium WebDriver was also performed to validate repetitive login scenarios. Automation helped in improving testing efficiency and ensured consistent execution of test cases.

Overall, the system meets most of the functional requirements related to login and performs reliably. The identified defects are manageable and can be resolved to further enhance system performance and user experience. The system is considered stable and ready for deployment after fixing the reported issues.

CHAPTER 8
CONCLUSION AND FUTURE SCOPE

8.1 CONCLUSION

The Water Jar Distribution System is developed to make the process of managing water jar supply easy, fast, and well organized. In the existing system, all work such as maintaining customer details, taking orders, managing deliveries, and tracking payments is done manually using notebooks or phone calls. This creates problems like errors, confusion, difficulty in tracking records, and wastage of time. To solve these issues, a fully working web-based system is developed that stores and manages all data in a proper and structured way.

The system allows customers to place orders easily, enter their details, and check their order status anytime. The supplier (admin) can manage customers, view and process orders, assign deliveries to delivery staff, and monitor overall system activities. Delivery staff can view assigned orders and update delivery status, which helps in proper tracking of deliveries. This improves coordination between all users and ensures smooth operation of the system.

The project is developed using modern technologies such as JavaScript, HTML, CSS, Bootstrap, and React.js for the frontend, while Node.js and Express.js are used for backend development. MongoDB is used as the database to store customer and order data. Tools like GitHub, Postman, and Visual Studio Code are used during development. The system runs on Windows 11 and provides a user-friendly interface.

Overall, the system reduces manual work, minimizes errors, saves time, and improves efficiency. It also helps in better record management, easy data access, and faster decision-making through reports. By integrating all operations into one platform, the project makes the water jar distribution process more reliable, transparent, and effective.

8.2 FUTURE SCOPE

1. **Online Payment Integration**

   This feature will allow customers to make payments through online methods such as UPI, debit/credit cards, or net banking. It will reduce the need for cash handling and make the payment process faster, secure, and more convenient.

2. **Mobile Application**

   A mobile app can be developed for Android/iOS so that users can easily place orders, track deliveries, and manage their accounts from their smartphones. This will improve accessibility and user experience.

3. **SMS and Email Notifications**

   The system can send automatic SMS or email notifications to customers for order confirmation, delivery updates, and payment reminders. This will keep users informed and improve communication.

4. **Real-Time Order Tracking**

   Customers can track their order and delivery status in real time on a map or through live updates, which will increase transparency and trust.

5. **Analytics and Reports**

   Advanced reports and analytics can be added to analyze sales, deliveries, and customer behavior, which will help in better decision-making.

CHAPTER 9

REFERENCES AND BIBLIOGRAPHY

9.1 REFERENCES

To understand water jar distribution challenges and practical software solutions, various project resources, tools, and technical documentation were studied.

1. Internal Project Repository and Source Code – Study of implemented modules for customer management, order processing, delivery management, and report generation.
2. API Testing and Backend Documentation – Review of backend APIs and request-response handling using tools like Postman for proper data communication.
3. Water Delivery System Study – Analysis of existing water supply methods to understand order handling, delivery process, and payment tracking.
4. MERN Stack Documentation – Use of official documentation for React.js, Node.js, Express.js, and MongoDB for system development.
5. Web Development References – Study of HTML, CSS, Bootstrap, and JavaScript for designing responsive and user-friendly web pages.

9.2 BIBLIOGRAPHY

- React Documentation: https://react.dev/
- Node.js Documentation: https://nodejs.org/docs/latest/api/
- Express.js Documentation: https://expressjs.com/
- MongoDB Documentation: https://www.mongodb.com/docs/
- Mongoose Documentation: https://mongoosejs.com/docs/